

EDUCATIVE COMMENTARY ON ISI 2017 MATHEMATICS PAPERS

(Last Revised on 31/03/2022)

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Introduction

The commentary on ISI B.Sc. Entrance Test for 2015 was written after those for 2016 and 2014. The quality of the questions in these two years had raised expectations. But the first paper of 2015 turned out to be very disappointing. Also, a look at the question papers prior to 2014 revealed certain common patterns which were repeated. So the element of novelty was not so strong as it was when the commentaries for 2016 and 2014 were written. To some extent such repetitions are to be expected, given the limited syllabus and the large number of questions to be asked. Instead of asking 30 questions in Paper 1 to be solved in two hours, it would be far better to ask only about 15 really good questions.

As a result, the plan to write exhaustive commentaries on ISI Entrance Tests was curtailed to only the four recent ones, viz. from 2014 onwards as these are the most relevant. A few students admitted through them could still be studying at ISI.

For reasons far from clear, the official version of the 2017 question papers has not been uploaded on the ISI website. The present commentary is based on unofficial versions of the two papers. Revisions will be made when the official versions are released.

As in the other commentaries, unless otherwise stated, all the references

are to the author's *Educative JEE Mathematics*, published by Universities Press, Hyderabad.

I am thankful to Deepanshu Rajvanshi who corrected an omission and several errors in an earlier draft of the present commentary. Readers are invited to send their comments and point out errors, if any, in the commentary. They may be sent either by e-mail (kdjoshi314@gmail.com) or by an SMS or a WhatsApp message on mobile (9819961036).

ISI BStat-BMath-UGA-2017 Paper with Comments

N.B. Each question has four options of which ONLY ONE is correct. There are 4 marks for a correctly answered, 0 marks for an incorrectly answered question and 1 mark for each unattempted question.

Q.1 Consider a quadratic equation $ax^2+2bx+c=0$ where a, b, c are positive real numbers. If the equation has no real roots, then which of the following is true?

- (A) a, b, c cannot be in AP or HP, but can be in GP.
- (B) a, b, c cannot be in GP or HP, but can be in AP.
- (C) a, b, c cannot be in AP or GP, but can be in HP.
- (D) a, b, c cannot be in AP, GP or HP.

Answer and Comments: (C). The connection with quadratic equations is superficial. The first statement is a fancy way of saying that $b^2 < ac$. The success of the solution lies in the observation that three numbers are in a particular type of a progression if and only if the middle one is the corresponding type mean of the other two. So, let A, G and H be the arithmetic, the geometric and the harmonic means respectively of a and c . Then $G^2 = ac$ and so the inequality $b^2 < ac$ implies $b^2 < G^2$ and hence, keeping in mind that all the numbers are positive, that

$$b < G \tag{1}$$

This proves that a, b, c cannot be in GP. We can similarly calculate A and H in terms of a and b and decide how they compare with b . But we can as well use the inequality that

$$H \leq G \leq A \tag{2}$$

The second part of this inequality is popularly called the A.M.-G.M. inequality. The first half is comparatively less known. It follows by applying the A.M.-G.M. inequality to the reciprocals $\frac{1}{a}$ and $\frac{1}{b}$.

In view of (1), (2) automatically rules out that $b = A$. So, a, b, c are neither in G.P. nor in A.P. Whether they are in H.P. or not depends

on whether $b = H$. There is nothing to rule this possibility out. So, by taking a, c to be any positive real numbers and setting $b = \frac{2ac}{a+c}$ (which is the harmonic mean of a and c), we see that it is possible for a, b, c to be in HP. ■

A good, simple problem based on the A.M.-G.H.-H.M. inequality, requiring very little computation once the essential idea strikes that three numbers are in a progression of a particular type if and only if the middle one is the corresponding type of mean of the other two.

The full A.M.-G.M.-H.M. inequality (3) rarely figures in applications. As noted above its second half is far more popular than the first one. Interestingly, there is one situation where all the three means figure and their inequality assumes a natural interpretation. And, even at the risk of a digression, it deserves to be mentioned. Surprisingly, it is from music. The arithmetic mean is more popularly called the average and we use it all the time in day to day life. The geometric mean of two numbers, say a and c has a geometric significance as the side of a square whose area equals that of a rectangle with sides a and c . Some old books mention that the term ‘harmonic mean’ comes from music. This is plausible because harmony of notes is very important in music. But rarely is the connection explained in detail.

In music, when two notes of different frequencies are played together, maximum harmony is reached when one of them is a multiple of the other. A well known fact in physics is that the frequency of sound produced by a vibrating string is inversely proportional to its length. So, if two strings of lengths L_1 and L_2 produce notes of frequencies f_1 and f_2 , then a string of length $\frac{L_1 + L_2}{2}$ will produce a frequency which equals their harmonic mean $\frac{2f_1f_2}{f_1 + f_2}$. If f_1 is taken as some base frequency, called *sa* or *shadja*, in Indian classical music, and $f_2 = 2f_1$, then f_2 is the frequency of the note called the upper *sa* or *taar shadja*. When the string is plucked at a point exactly at the middle, it produces the frequency $\frac{4}{3}f_1$ which is simply the harmonic mean of f_1 and $2f_1$. It is called the *madhyam* or more precisely the *shudh madhyam* while the note whose frequency is the A.M. $\frac{3}{2}f_1$ is called the *pancham*. Thus, in

the musical octave, the *pancham* is sharper than the *madhyam* because of the A.M.-H.M. inequality.

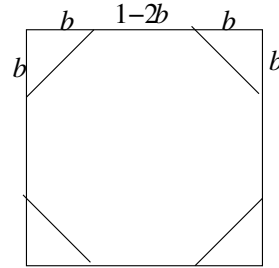
What about the geometric mean $\sqrt{2}f_1$? $\sqrt{2}$ is an irrational number. But the frequency $\sqrt{2}f_1$ comes very close to the note called the *teevra madhyam*. Summing up, the *madhyam*, the *teevra madhyam* and the *pancham* are, respectively, the harmonic, the geometric and the arithmetic means of the *shadja* and the *taar shadja*. And the musical interpretation of the inequality is that the *pancham* is sharper than the *teevra madhyam*, which, in turn, is sharper than the *shudh madhyam*.

Q.2 A unit square has its corners chopped off to form a regular polygon with eight sides. What is the area of this polygon?

- (A) $2(\sqrt{3} - \sqrt{2})$ (B) $2\sqrt{2} - 2$ (C) $\frac{\sqrt{2}}{2}$ (D) $\frac{7}{9}$

Answer and Comments: (B). A good diagram is a key to understand the problem as well as to solve it.

It is not given that the corners are of equal size or even that they are isosceles triangles. But such things are to be assumed in the very concept of a corner. So let b be the side of the corner lying along an edge of the square. Then we get a diagram as shown.



Equating the portion of the side left with the hypotenuse of the corner, we get

$$1 - 2b = \sqrt{2}b \quad (1)$$

Solving,

$$b = \frac{1}{2 + \sqrt{2}} \quad (2)$$

We now get the length of the side of the octagon as $\frac{\sqrt{2}}{2 + \sqrt{2}}$. There is a formula for the area of a regular n -gon in terms of the length of its

side and using it we can answer the question. But that would be too elaborate. In the present case, the octagon is the figure that results by removing the four corners from the unit square. So its area, say A , is obtained by subtracting the removed area which is simply $2b^2$ from 1. Hence,

$$\begin{aligned} A &= 1 - \frac{2}{(2 + \sqrt{2})^2} \\ &= 1 - \frac{2}{6 + 4\sqrt{2}} = 1 - \frac{1}{3 + 2\sqrt{2}} \\ &= \frac{2 + 2\sqrt{2}}{3 + 2\sqrt{2}} \end{aligned} \tag{3}$$

As it stands, this answer does not match with any of the given options. To remedy this, we rationalise the denominator. Thus

$$\begin{aligned} A &= \frac{(2 + 2\sqrt{2})(3 - 2\sqrt{2})}{(3 + 2\sqrt{2})(3 - 2\sqrt{2})} \\ &= \frac{(2 + 2\sqrt{2})(3 - 2\sqrt{2})}{9 - 8} \\ &= -2 + 2\sqrt{2} \end{aligned} \tag{4}$$

which matches with (B). ■

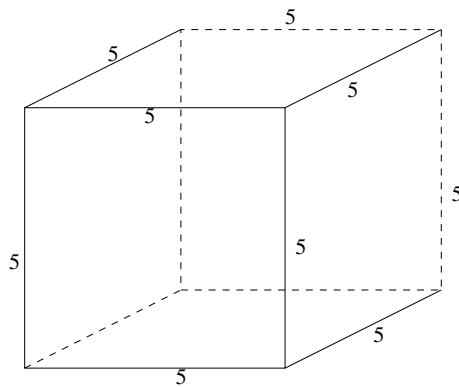
A straightforward problem, but not without some room for elegance. Had we applied the formula for the area of a regular n -gon with each side d , viz., $n \times \frac{d^2}{4} \cot(\pi/n)$, we would have got the same answer by putting $n = 8$ and $d = \sqrt{2}b = \frac{\sqrt{2}}{2 + \sqrt{2}}$. But we would have to calculate $\cot(\pi/8)$ i.e. $\cot(22.5^\circ)$ which is not a very standard figure. Instead, we got the answer effortlessly by simply subtracting the removed area. The discrete analogue of this technique is called **complementary counting** (see Comment No. 1 of Chapter 1).

A little work in rationalising a surd is also needed at the end. But it is not very laborious and so this problem is a reasonable one.

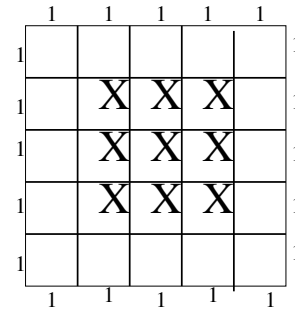
- Q.3 A solid cube of side five centimeters has all its faces painted. The cube is sliced into smaller cubes each of side one centimeter. How many of these cubes will have paint on exactly one of its faces?

- (A) 25 (B) 54 (C) 126 (D) 150

Answer and Comments: (B). Another simple geometric problem where half the success lies in visualising it correctly. We show the cube in (a) of the figure below. In all the cube gets divided into $5^3 = 125$ smaller cubes. Out of these, $3^3 = 27$ will lie in the interior of the cube and none of their faces will be painted. For each of the remaining 98 smaller cubes, at least one face will lie along a face of the bigger cube and hence will get painted. We want the number of those smaller cubes which have exactly one face lying along a face of the original cube.



(a)



(b)

In (b) of the figure we show one face of the original cube divided into 25 squares of side 1 each. Out of these 9 lie in the interior while 16 lie along the boundary. The latter will also have another face lying along some other face of the cube and hence is painted. So there are only 9 smaller cubes which have exactly one face painted. They are different for the different faces of the bigger cube. As the big cube has six faces, the number of smaller cubes having exactly one face painted is $9 \times 6 = 54$. ■

Another good, simple problem. It would have been more interesting to ask the number of smaller cubes which have exactly two of their faces painted. On each face, these are the ones which appear on the border and there are 16 of them (those that are not marked in part (b) of the figure). So, it is tempting to think that in all there are 16×6 smaller cubes which have two of their faces painted. But in doing so,

we are counting each cube twice because it lies along an edge which is the common boundary of two adjacent faces. So, the correct answer is not 96 but only 48. But even this is not correct, because cubes which are at the eight vertices of the big cube have three of their faces painted. Together they count for 24 smaller painted squares. So we need to subtract 24 from 96 and *then* divide by 2. That gives 36. A slightly easier approach is to count only the small cubes that lie along an edge but not at either of its two ends. There are three such small cubes along each edge. As there are twelve edges, the number of small cubes which have exactly two of their faces painted is 36. Summing up, each of the 125 smaller cubes can have at most three of its faces painted. The number of small cubes with 0, 1, 2 and 3 faces painted is, respectively, 27, 54, 36 and 8. Their total is 125 as expected. Also the total number of painted faces (of smaller cubes) is $27 \times 0 + 54 \times 1 + 36 \times 2 + 8 \times 3 = 54 + 72 + 24 = 150$, also as expected because each of the six faces of the big cube contains 25 faces of the small cubes.

Q.4 Let z be a complex number such that $\frac{z-i}{z-1}$ is purely imaginary. Then the minimum value of $|z - (2 + 2i)|$ is

- (A) $2\sqrt{2}$ (B) $\sqrt{2}$ (C) $\frac{3}{\sqrt{2}}$ (D) $\frac{1}{\sqrt{2}}$

Answer and Comments: (B). Let us first interpret what is asked. Geometrically, $|z - (2 + 2i)|$ is the distance of the point z from the fixed point $2 + 2i$ in the complex plane. The point z is a variable point which satisfies a certain given condition. If we can interpret this condition suitably, it will translate into the equation of some curve, say C , in the complex plane. Once we identify this curve C , the problem reduces to finding the point on it which is closest to the point $2 + 2i$. This is an easy task if C comes out to be a straight line or a circle. Otherwise it is, by itself, another problem. (For example, if C comes out to be a conic, we shall have to take its parametric equations with some parameter t , express the distance from the point $2 + 2i$ as a function of t and then minimise it.)

So, let us first identify this curve C on which z moves if it satisfies the

condition that the ratio $\frac{z-i}{z-1}$ is a purely imaginary number. A brute force method in dealing with problems about complex numbers is to resolve them into real and imaginary parts. In the present problem, for example, if we write $z = x + iy$ as usual, then the given condition means

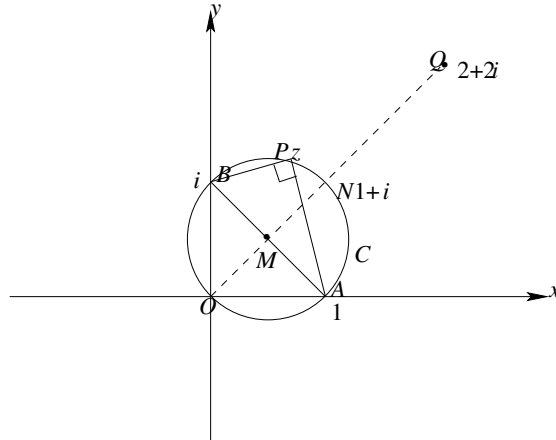
$$\frac{x + i(y-1)}{(x-1) + iy} = ib \quad (1)$$

for some real number b . Multiplying both the sides by the complex conjugate of the denominator,

$$(x + i(y-1))((x-1) - iy) = ib((x-1)^2 + y^2) \quad (2)$$

Equating the real parts of both the sides as well as their imaginary parts will give a system of two equations in the variables x and y with b as a parameter. Eliminating b between these two equations will give us the equation of the curve C .

It is obvious that method will entail considerable algebraic manipulations. But there is a way to interpret the data geometrically. Let $z, 1, i$ and $2 + 2i$ represent the points P, A, B and Q in the Argand diagram. Then the complex numbers $z - i$ represents the line segment from B to P . Similarly, the complex number $z - 1$ represents the segment from A to P . The condition that the ratio of these two numbers is purely imaginary is equivalent to these two segments being perpendicular to each other as shown in the figure below.



Since $\angle APB = 90^\circ$, the locus of P is the circle with AB as a diameter and hence $M = \frac{1+i}{2}$ as its centre. We have called it C . Since it is a circle, the point, say N , on it which is closest to the point $Q = 2 + 2i$ is the closer end of the diameter passing through Q . As the centre of C is the point $\frac{1+i}{2}$, it is clear that N is the point $1 + i$. Its distance from Q is $|1 + i|$ i.e. $\sqrt{2}$. ■

Another simple problem once the characterisation of perpendicularity of two segments in terms of complex numbers strikes. The paper setters have been careful to give the numerical data in such a way that the closest point N to Q can be identified merely by inspection. When a problem is designed to test one particular idea, it is unfair to complicate it with laborious calculations not relevant to its main theme.

Q.5 Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that for any two real numbers x and y ,

$$|f(x) - f(y)| \leq 7|x - y|^{201}$$

Then

- | | |
|---------------------------|---------------------------|
| (A) $f(101) = f(202) + 8$ | (B) $f(101) = f(201) + 1$ |
| (C) $f(101) = f(200) + 2$ | (D) None of the above. |

Answer and Comments: (D). The problem is poorly designed. The choice of the particular figures 201 and 7 is highly arbitrary to say the least. One wonders if they were obtained by splitting 2017, the calendar year in which the test is held!

More seriously, the condition which the function f is given to satisfy trivially implies that it is continuous. Actually, more is true as will be proved later. So it is superfluous to give continuity of f .

The options given are matchingly silly. Any constant function meets all the conditions in the data. So that rules out (A), (B) and (C). Hence (D) is correct. ■

A slightly non-trivial problem would have been to ask to show that under the given conditions, $f(x)$ has to be a constant. This can be

done as follows. Fix any $x \in \mathbb{R}$. Then for any $h \neq 0$, we have

$$0 \leq \left| \frac{f(x+h) - f(x)}{h} - 0 \right| \leq 7|h|^{200} \quad (1)$$

as we see by taking $y = x + h$ in the given condition. As $h \rightarrow 0$, the R.H.S. tends to 0 and hence so does the L.H.S. by the popular **Sandwich Theorem** for limits. (Actually the redundant left half of (1) was added only to make the Sandwich Theorem applicable. After all, you need two pieces of bread to make a sandwich!)

Thus we have proved that $f(x)$ is differentiable everywhere and further that its derivative vanishes identically. Concluding that f is a constant function from this is often taken as obvious by those who confuse a statement with its converse. The converse, viz. that the derivative of a constant function vanishes everywhere is indeed trivial. But to go the other way, you need Lagrange's Mean Value Theorem. For any two a, b with $a < b$, there is some $c \in (a, b)$ such that $f(b) - f(a) = f'(c)(b - a)$. Since $f'(c) = 0$ regardless of where c lies, we have $f(b) = f(a)$. So, f is a constant.

Note that the constant 7 and the exponent 201 have little role here. Suppose, more generally, that $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies

$$|f(x) - f(y)| \leq A|x - y|^\alpha \quad (2)$$

where A and α are some positive constants. Then the argument above will apply equally well if $\alpha > 1$. For $0 < \alpha < 1$, little can be concluded beyond continuity of $f(x)$. The case $\alpha = 1$ is most interesting. It is tempting to think that in this case $f(x)$ must be of the form $\pm Ax + B$ for some constant B . But there are other examples too which we shall not go into.

It is entirely possible that there is some hidden agenda in asking this problem. Perhaps the idea is to compare the performance on this question with that on the rest. If candidates who do well elsewhere fail in this problem, there is a cause for concern for their training.

- Q.6 In the Mathematics department of a college, there are 60 first year students, 84 second year students and 108 third year students. All of these students are to be divided into project groups such that each

group has the same number of first year students, the same number of second year students and the same number of third year students. What is the smallest possible size of each group?

- (A) 9 (B) 12 (C) 19 (D) 21

Answer and Comments: (D). Let n be the number of groups. All 60 students of the first year have to be distributed equally into these n groups. So 60 is a multiple of n , or equivalently, n divides 60. Similarly it divides 84 and 108. So it is a common divisor of 60, 84 and 108. The largest value we can take is their g.c.d. (greatest common divisor). By inspection $n = 12$. To finish, we divide the total number of students, $60 + 84 + 108$ by 12 to get 21 as the smallest possible size of each group.

■

It takes more time to read the problem than to work it out. The only ‘thoughtful’ step is to realise that the number of groups is a common divisor of 60, 84 and 108. After that, it is a problem of elementary school. There they fit in nicely, because one of the criteria of good education is the ability to apply it to real life problems. An alarmingly large percentage of college entrants simply dread ‘word problems’. Perhaps it is to prevent their entry that such a simple problem is designed, which properly belongs as a puzzle in the recreation section of a newspaper. (A similar remark was made about Q.20 in Paper 1 of 2015. But that problem required at least some deductive reasoning. The present one requires little more than finding the g.c.d. of three numbers which are from the multiplication table of 12.)

Looking at the social aspects of the data, it would have been better if the figures 60, 84 and 108 stood for students enrolled for degree programs in Mathematics, Physics and Chemistry respectively. That will be a realistic reflection of the relative popularity of these subjects. As the problem stands, one gets the impression that the Mathematics program of the college is declining in popularity!

Q.7 Let a, b, c be real numbers each greater than 1 such that

$$\frac{2}{3} \log_b a + \frac{3}{5} \log_c b + \frac{5}{2} \log_a c = 3$$

If the value of b is 9, then the value of a must be

- (A) $\sqrt[3]{81}$ (B) $\frac{27}{2}$ (C) 18 (D) 27

Answer and Comments: (D). Like Q.2 in Paper 1 of 2014, the present problem involves logarithms w.r.t. different bases. In such problems, it is convenient to express all such logarithms in terms of logarithms w.r.t. some common base. The choice of this common base is unimportant and it need not even be specified. We take this common base as 3 because we are given that the value of b is 9 which is a power of 3. We now write $\log_3$ simply as $\log$. Denote $\log a, \log b$ and $\log c$ by x, y, z respectively. Then by the formula for the change of base,

$$\log_b a = \frac{\log a}{\log b} = \frac{x}{y}, \log_c b = \frac{y}{z} \text{ and } \log_a c = \frac{z}{x} \quad (1)$$

Hence the given equation becomes

$$\frac{2x}{3y} + \frac{3y}{5z} + \frac{5z}{2x} = 3 \quad (2)$$

We are given the value of b as 9 and hence

$$y = \log 9 = 2 \quad (3)$$

So, effectively, (2) is an equation in two unknowns x and z . In general such an equation will have infinitely many solutions and so it is impossible to determine x and hence a uniquely from it. A single equation in several unknowns has a unique solution only in degenerate cases. A simple example is the equation $u^2 + v^2 = c$ in the unknowns u and v . For $c > 0$ it has infinitely many solutions while for $c < 0$ it has no (real) solutions. But for the degenerate case $c = 0$, it has a unique solution $u = 0, v = 0$.

So the only thing we can hope for is that (2) falls in some degenerate or extreme case whereby it has a unique solution. Let us write the three terms on the L.H.S. as u, v, w respectively. Then (2) says that their A.M. is 1. But by a direct calculation, $uvw = 1$ and hence the G.M. of u, v, w is also 1. So this is an extreme case where the A.M. of three positive real numbers equals their G.M. This can happen only

when all the numbers are equal and hence each one of them is 1 here. So now we get three separate equations:

$$2x = 3y, 3y = 5z \text{ and } 5z = 2x \quad (4)$$

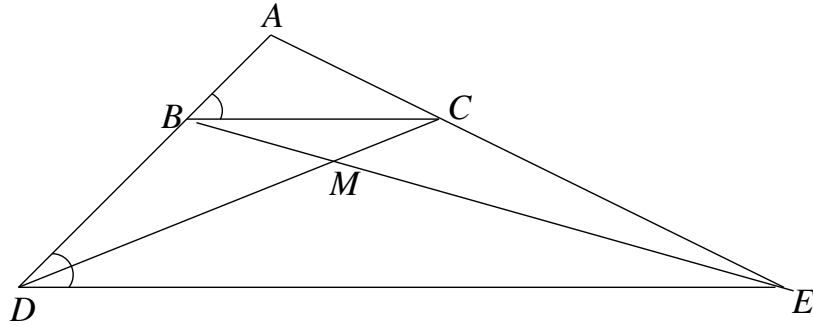
As $y = 2$, the first equation gives $x = 3$ and hence $a = 3^x = 3^3 = 27$. ■

The problem is a combination of the worn out A.M.-G.M. inequality and some elementary facts about logarithms. The former already appeared in Q.1 above. The latter permits only a limited variety of problems. So the present problem, although a challenge for the untutored, is a dull one for those who are familiar with such problems.

Q.8 Consider a triangle ABC . The sides AB and AC are extended to points D and E respectively, such that $AD = 3AB$ and $AE = 3AC$. Then one diagonal of $BDEC$ divides the other diagonal in the ratio

- (A) $1 : 3$ (B) $1 : \sqrt{3}$ (C) $1 : 2$ (D) $1 : \sqrt{2}$

Answer and Comments: (A). The data is pictured in the figure below where the diagonals of the quadrilateral $BDEC$ intersect at the point M .



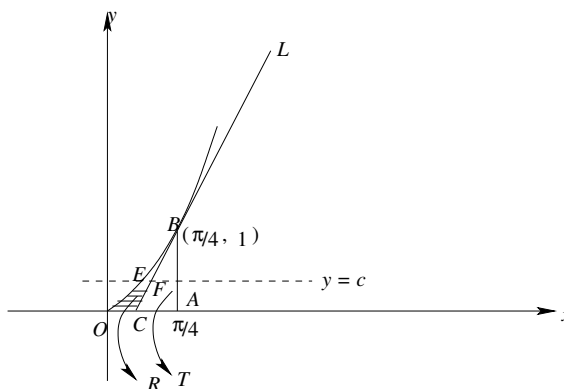
Since $AD : AB = AE : AC = 3$, the triangles ADE and ABC are similar and $DE : BC = 3$. We also get that BC and DE are parallel to each other. This makes the triangles BMC and EMD similar. Hence both $BM : ME$ and $CM : MD$ are $BC : DE = 1 : 3$. ■

An absolutely simple problem based on similar triangles.

Q.9 The area of the region bounded by the curve $y = \tan x$, the x -axis and the tangent to the curve at $x = \frac{\pi}{4}$ is

- (A) $\log_e 2 - \frac{1}{2}$ (B) $\log_e 2 + \frac{1}{2}$ (C) $\frac{1}{2}(\log_e 2 - \frac{1}{2})$ (D) $\frac{1}{2}(\log_e 2 + \frac{1}{2})$

Answer and Comments: (C). A straightforward problem about calculating the area below the graph of a function $y = f(x)$. The only catch here is that the vertical boundaries (which correspond to the end points of the interval over which $f(x)$ is to be integrated) are not given. Also the lower boundary of the region, say R , is partly the x -axis and partly the tangent at the point $(\frac{\pi}{4}, 1)$ to the curve $y = \tan x$. So, before we carry out the integration, we have to look for this missing information and the best way is to sketch the region R as we do in the figure below. The tangent to the curve $y = \tan x$ at the point $B = (\pi/4, 1)$ is shown by the line L . It cuts the x -axis at the point C . The region R is bounded above by the curve $y = \tan x$ and below by the segments OC and CB . It is shown by shading by horizontal segments.



The upper boundary of the region R is an arc of a single curve $y = \tan x$ from $O = (0, 0)$ to $B = (\frac{\pi}{4}, 1)$. But the lower boundary consists of two parts, the segments OC and CB . To evaluate the area of R , we can proceed in three ways listed below,

- (i) We can divide the region R into two subregions, one lying on the left of the vertical line through the point C and the other to its right. For this, we shall first have to identify the point C and for this we shall first have to find the equation of the line L . The area

of each subregion will have to be calculated by an appropriate integral.

- (ii) Instead of splitting the region R , we add to it the triangle ABC (shown as T in the figure) to get a bigger region S , which is bounded above by the curve $y = \tan x$ and below by the segment OA . We calculate the area of S and subtract from it the area of the triangle T .
- (iii) We can evaluate the area of R by horizontal slicing. A typical horizontal line $y = c$ intersects the region R in a segment EF where $E = (\tan^{-1} c, c)$ and F will come out after we find the equation of the line L . If this point is $(g(c), c)$, then the area of R will be $\int_0^1 (g(y) - \tan^{-1} y) dy$.

In the first two approaches, we have to identify the point C . For this we first need the equation of the line L , i.e. the tangent to the curve $y = \tan x$ at the point $(\frac{\pi}{4}, 1)$. Since $\frac{d}{dx}(\tan x) = \sec^2 x$, the slope of L is $\sec^2(\pi/4) = 2$. And so its equation is

$$y = 1 + 2(x - \frac{\pi}{4}) \quad (1)$$

Hence

$$C = (\frac{\pi}{4} - \frac{1}{2}, 0) \quad (2)$$

It is now instructive to compare the merits and demerits of the three methods above and choose the most convenient one.

- (i) For the first approach, we shall have to calculate two integrals $\int_0^{\pi/4-1/2} \tan x \, dx$ and $\int_{\pi/4-1/2}^{\pi/4} (\tan x - 1 - 2(x - \frac{\pi}{4})) dx$.
- (ii) Here we shall have to calculate the integral $\int_0^{\pi/4} \tan x \, dx$ and subtract from it the area of the triangle T .
- (iii) From (1), the function $g(y)$ comes out as $\frac{\pi}{4} + \frac{y-1}{2}$. The area of the region R is then the integral $\int_0^1 (\frac{\pi}{4} + \frac{1-y}{2} - \tan^{-1} y) dy$.

There is not much conceptual difference between (i) and (ii). If we split the second integral, viz. $\int_{\pi/4-1/2}^{\pi/4} (\tan x - 1 - 2(x - \frac{\pi}{4}))dx$ into two parts by splitting the integrand and add the first part to the first integral, viz. $\int_0^{\pi/4-1/2} \tan x dx$, then by the additivity property of the integrals we get the integral $\int_0^{\pi/4} \tan x dx$ which is precisely the integral in (ii). The integral of the remainder is the area of the triangle T . The third approach, however, is qualitatively different. Here we get the answer by a single integration. But we shall have to find an antiderivative for $\tan^{-1} y$. This can be done using integration by parts. But that means we shall be spending what we saved.

This often happens in mathematics. No matter which approach you take, the work involved remains essentially the same. It is only in rare cases that we can bypass some arduous work by cleverly transforming the problem. In the solution to Q.2, we mentioned complementary counting as an example of such a transformation. In the present problem, the second approach resembles complementary counting because the area of the triangle T can be found without integration. Its base is $\frac{\pi}{4} - (\frac{\pi}{4} - \frac{1}{2})$ which is simply $\frac{1}{2}$ while its height is 1. So the area of the triangle T is $\frac{1}{4}$. We need to subtract this from the area of S which is the integral $\int_0^{\pi/4} \tan x dx$. Hence the area of the region R is given by

$$\begin{aligned} A(R) &= \int_0^{\pi/4} \tan x dx - \frac{1}{4} \\ &= \log_e(\sec x) \Big|_0^{\pi/4} - \frac{1}{4} = \log_e \sqrt{2} - \log_e 1 - \frac{1}{4} \\ &= \frac{1}{2} \log_e 2 - 0 - \frac{1}{4} = \frac{1}{2}(\log_e 2 - \frac{1}{2}) \end{aligned} \tag{3}$$

which tallies with (C). ■

A straightforward problem, but the work needed is time consuming. But then perhaps that balances Q.5 (where no work is needed) and Q.6 (where the only work needed is to find the g.c.d. of 60, 84 and 108). So the overall package deal is not bad. But since all questions

carry equal marks and it is next to impossible to finish all the questions in the given time, those who pick the less demanding problems stand to gain. Exactly the same problem has been asked in JEE 1988. See Exercise (17.3)(ix). Admittedly, most problems on finding areas are similar. Still such an outright duplication should have been avoided.

- Q.10 Let V be the set of vertices of a regular polygon with twenty sides. Three distinct vertices are chosen at random from V . Then the probability that the chosen triplet of vertices forms a right angled triangle is

(A) $\frac{7}{19}$ (B) $\frac{1}{19}$ (C) $\frac{3}{38}$ (D) $\frac{1}{38}$

Answer and Comments: (B). Probability theory is not in the ISI syllabus. But the present problem, although superficially a probability problem, is really two counting problems put together, because the desired probability, say p , is the ratio of two numbers given by

$$p = \frac{|F|}{|S|} \quad (1)$$

where S stands for the set of all possible outcomes and F stands for the subset of F consisting of the outcomes that are ‘favourable’, i.e. those outcomes in which the given event occurs. Here $|X|$ stands for the number of elements in a set X . The set S is often given a fancy name, the **sample space**. This simple minded definition of probability is applicable only when the sample space is finite and, more importantly, when all elements of it are equally likely. This is the meaning given to expressions like ‘at random’.

In the present problem, the sample space S is the set of all ways to select three elements of the vertex set V . As $|V|$ is given as 20, we get

$$|S| = \binom{20}{3} = \frac{20 \times 19 \times 18}{3!} = 19 \times 60 \quad (2)$$

Note that we have not multiplied 19 and 60. To get p , we have to divide by $|S|$ and at that time the factorisation 19×60 will be more convenient, especially so since in all the four options given, the denominators are multiples of 19. So it is foolish to waste precious time in multiplying 60

and 19 prematurely and also thereby increase the chances of a numerical slip.

Now comes the real task of finding $|F|$, the number of favourable cases. In the present problem, this is the set of those triples $\{A, B, C\}$ in which A, B, C are vertices of a right angled triangle. As the order is unimportant, we suppose that ABC is right angled at B . We are given that the angular distance between any two consecutive vertices is the same, say, α . (Actually, $\alpha = \frac{2\pi}{20} = \frac{\pi}{10} = 18^\circ$. But that is not crucial here.) Hence the angular distance between any two (distinct) vertices will be a multiple of α , ranging from α to 10α .

Consider a triangle ABC right angled at B . Without loss of generality, assume that the side AB is smaller than (or possibly equal to) the side BC . Then the angular distance between A and B is $n\alpha$ for some $n = 1, 2, 3, 4$ or 5 . For each n , there are twenty such segments whose end points are at an angular distance $n\alpha$. Each segment determines two possible locations for the third vertex C . (These two positions along with A and B form a rectangle.) So, for every n , there are 40 right angled triangles in which the shorter side has angular length $n\alpha$. However, for $n = 5$, AB and BC are equal in length and so the same triangle gets counted twice. So the total count of triangles is not 40×5 but $40 \times 4 + 20 = 180$. So the favourable set F has 180 elements. Hence by (1) and (2), $p = \frac{180}{19 \times 60} = \frac{3}{19}$. ■

Although the solution above is reasonably simple, there is a more elegant way to count $|F|$. For ABC to be right angled, the longest side must be a diameter. There are in all 10 diameters whose ends are from V . For each such diameter, the third vertex can be any of the remaining 18 vertices. So, $|F| = 10 \times 18 = 180$.

The second solution is based on the geometric result that the angle in a semi-circle is a right angle. As the solution to Q.4 also hinged on this idea, there is some duplication.

Q.11 A “basic row operation” on a matrix means adding a multiple of one row to another row. Consider the matrices

$$A = \begin{pmatrix} x & 5 & x \\ 1 & 3 & -2 \\ -2 & -2 & 2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 0 & 21 \\ 1 & -1 & -14 \\ 0 & \frac{4}{3} & 4 \end{pmatrix}$$

It is given that B can be obtained from A by finitely many basic row operations. Then, the value of x is:

- (A) 3 (B) -3 (C) -1 (D) 2

Answer and Comments: (A) This is one of those problems where the success depends upon whether the statement of the problem rings a familiar bell in some way. What is called a “basic row operation” is more customarily called an elementary row operation. But whatever be the name, the technique of adding a multiple of one row to another is familiar as a tool to simplify a determinant. Under each such operation, the value of the determinant remains unchanged. So the same is true if one matrix is obtained from the other by applying any (finite) number of basic row operations. Here A and B are given matrices. Denote their determinants by $|A|$ and $|B|$ respectively. Then $|B| = |A|$. The latter can be calculated directly as

$$|B| = 21 \times \frac{4}{3} = 28 \quad (1)$$

$|A|$ comes out to be a polynomial in the indeterminate x .

$$|A| = 2x + 5(4 - 2) + 4x = 6x + 10 \quad (2)$$

Equating $|A|$ with $|B|$ gives $6x = 18$ and hence $x = 3$. ■

Another simple problem, once the key idea strikes. And the key idea is not at all foreign. But it is one thing to know a fact. It is quite another to realise where it will be applicable. In the elegant solution to the last problem, the key fact needed was that the angle in a semi-circle is a right angle. It is easy to hit this if you start with a semi-circle. But it is not so easy to hit it if you start with a right angled triangle. In the present problem, however, the moment somebody sees the words ‘adding a multiple of one row to another’ the bell ought to ring that the determinant is invariant. If it doesn’t, there is no other way to solve the problem. In the last question, on the other hand, even if you missed the short cut, the answer can be obtained by other means. This is probably the dividing line between an elegant solution and a tricky one. An elegant solution does not preclude other, possibly

more laborious approaches. In fact its elegance stands out more in comparison to them. In a tricky solution, however, if the trick eludes you, you are helpless.

Q.12 Let C be a circle of area A and centre at O . Let P be a moving point such that its distance from O is always twice the length of a tangent drawn from P to the circle. Then the point P moves along

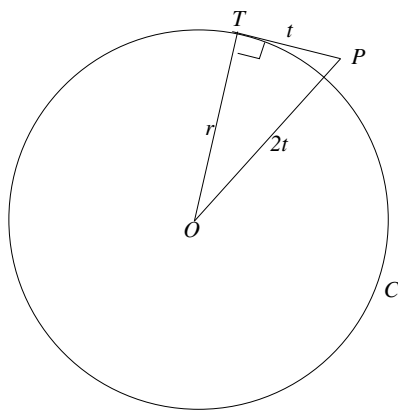
- (A) the sides of a square centred at O , with area $\frac{4}{3}A$
- (B) the sides of an equilateral triangle centred at O , with area $4A$
- (C) a circle centred at O , with area $\frac{4}{3}A$
- (D) a circle centred at O , with area $4A$

Answer and Comments: (C). The area of the circle C is given as some constant A . It would have been more natural, instead, to give its radius as some constant, say r . Of course, the two are related by

$$A = \pi r^2 \quad (1)$$

But the radius is a more directly appealing attribute of a circle and it is not clear why this twist was given.

So, let us take r as the radius of the circle C . Let PT be a tangent from P to C . Suppose $PT = t$. We are given that $OP = 2t$.



From the right angled triangle OTP we get

$$r^2 + t^2 = 4t^2 \quad (2)$$

So $t^2 = \frac{1}{3}r^2$. As a r is a constant so is t and hence $2t$. Therefore P traces a circle of radius $2t$ centred at O . Its area is $4\pi t^2 = \frac{4}{3}\pi 3t^2 = \frac{4}{3}\pi r^2 = \frac{4}{3}A$.

■

An absolutely simple problem requiring nothing beyond the perpendicularity of the tangent and the radius and the theorem of Pythagoras.

- Q.13 A moving line intersects the lines $x + y = 0$ and $x - y = 0$ at the points A and B such that the triangle with vertices $(0, 0)$, A and B has a constant area C . The locus of the midpoint of AB is given by the equation

$$\begin{array}{ll} \text{(A)} (x^2 + y^2)^2 = C^2 & \text{(B)} (x^2 - y^2)^2 = C^2 \\ \text{(C)} (x + y)^2 = C^2 & \text{(D)} (x - y)^2 = C^2 \end{array}$$

Answer and Comments: (B). This problem is of a similar spirit as the last one. But the last one was a pure geometry problem. In the present one, the data as well as the conclusion involve coordinates.

One of the advantages of coordinates is that many times it is unnecessary to draw a diagram. We can simply translate the data in terms of algebraic equations and then manipulate it. So, in the present problem, denote $(0, 0)$ by O . Let the midpoint of AB be $P = (h, k)$. We take A as $(a, -a)$ for some a and B as (b, b) for some b . Here a, b are two real parameters. We shall first express h and k in terms of these two parameters and then eliminate them with some additional piece of data to get the locus of P .

The first part is easy. Since $P = (h, k)$ is the midpoint of AB , we have

$$a + b = 2h \quad (1)$$

$$\text{and } b - a = 2k \quad (2)$$

The lines $x + y = 0$ and $x - y = 0$ intersect at O are at right angles. So, AOB is a right angled triangle. Equating its area with C gives

$$|OA||OB| = 2C \quad (3)$$

Squaring and noting that $|OA|^2 = 2a^2$ while $|OB|^2 = 2b^2$, we get

$$a^2b^2 = C^2 \quad (4)$$

We get the locus of P by eliminating a and b from (1), (2) and (4). Solving (1) and (2) for a and b ,

$$b = h + k \quad (5)$$

$$\text{and } a = h - k \quad (6)$$

Putting these into (4) and replacing h, k by x, y we get the locus of P as

$$(x + y)^2(x - y)^2 = C^2 \quad (7)$$

i.e. $(x^2 - y^2)^2 = C^2$. ■

the standard method to obtain the locus of a point P moving in the plane subject to some condition is to express its coordinates (often called **current coordinates** and usually denoted by (h, k) rather than by (x, y) which may be involved in some other equations in the data) in terms of a single parameter, and then eliminate this parameter using the condition which P is supposed to satisfy. The choice of the parameter is ours. But when the problem itself contains some 1-parameter family, the most natural choice is to take the parameter in such a way that as it varies we get different members of the given family. In the present problem, this family is the family of all lines through the point (h, k) . The equation of a typical member of this family is of the form

$$y - k = m(x - h) \quad (8)$$

In other words, we are taking the slope of the moving line as a parameter. Solving (8) simultaneously with $x + y = 0$ and $x - y = 0$ we get the points A and B . They come out to be

$$A = \left(\frac{mh - k}{m + 1}, \frac{k - mh}{m + 1} \right) \quad (9)$$

$$\text{and } B = \left(\frac{mh - k}{m - 1}, \frac{mh - k}{m - 1} \right) \quad (10)$$

respectively. As $P = (h, k)$ is the midpoint of AB , we get

$$\begin{aligned} 2h &= (mh - k)\left(\frac{1}{m+1} + \frac{1}{m-1}\right) \\ \text{and } 2k &= (mh - k)\left(\frac{1}{m-1} - \frac{1}{m+1}\right) \end{aligned} \quad (11)$$

both of which simplify to

$$h = mk \quad (12)$$

As a result, we have

$$|OA|^2 = 2 \frac{(mh - k)^2}{(m+1)^2} = 2 \frac{(h^2 - k^2)^2}{(h+k)^2} = 2(h-k)^2 \quad (13)$$

$$\text{and } |OB|^2 = 2 \frac{(mh - k)^2}{(m-1)^2} = 2 \frac{(h^2 - k^2)^2}{(h-k)^2} = 2(h+k)^2 \quad (14)$$

Putting these into (3) will give us the same locus as before. But this derivation is considerably more complicated than the earlier one which was based on two parameters a and b instead of a single parameter m . As a general rule, one should minimise the number of parameters. But as this problem shows, sometimes a clever choice of several parameters simplifies the work.

Q.14 Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{a, b, c, d, e\}$. How many functions $f : A \rightarrow B$ are there such that for every $x \in A$, there is one and exactly one $y \in A$ with $y \neq x$ and $f(x) = f(y)$?

- (A) 450 (B) 540 (C) 900 (D) 5400

Answer and Comments: (C). This question is of the same spirit as Q.2 of Paper 1 of ISI 2015 which asked for the number of all one-to-one functions from a set with 3 elements to a set with 6 elements. But the latter was utterly trivial, being a direct application of the product rule for counting. In the present problem, we have to do some thinking.

Let F be the set of all functions from A to B which satisfy the given condition. If $f \in F$, then by very design, f cannot be one-to-one. Still its failure to be a one-to-one function is well regulated. For example, it cannot map three elements of A to the same element of B .

The condition on f means that the six elements 1, 2, 3, 4, 5 and 6 of the domain are paired off into three pairs and f takes the same value on the two elements of the same pair, but different value on those of the other pairs. (Often such function is called a **two-to-one** function.) For example, consider the pairing $\{1, 5\}, \{2, 3\}, \{4, 6\}$ of A . Then f is uniquely determined by the values $f(1), f(2)$ and $f(4)$, 1, 2, 4 being the representatives of the three pairs. But these three values must be different. So, for each pairing, f is like a one-to-one function from a set with 3 elements (viz., the set of the three pairs in that pairing) to the set $B = \{a, b, c, d, e\}$. Now this problem is very similar to the utterly trivial problem mentioned above. The value on the first pair can be chosen in 5 ways. For each such choice the value on the second pair can be chosen in 4 ways and continuing, that on the third in 3 ways.

So, for each pairing of the domain set, we have $5 \times 4 \times 3 = 60$ elements in F . We now only need to multiply this figure by the number of possible pairings. In any pairing, the ‘mate’ of 1 can be chosen in 5 ways. Once selected, the remaining 4 elements can be paired in 3 different ways.

Summing up, there are 15 possible pairings and for each pairing there are 60 functions in F . So the total count is $15 \times 60 = 900$. ■

A good problem, certainly far better than the 2015 question which is absolutely unbecoming for a reputed institute. The reasoning above looks long because we have given it in detail. But it does not take much time to conceive it mentally and thereafter the calculations involved are minimal.

The problem of pairing an even number, say $2n$ of objects into n mutually disjoint pairs is interesting. Denote this number by a_n . Then it is the number of ways $2n$ participants in a tournament can be paired for the first round of matches. Clearly, $a_1 = 1$ and $a_2 = 3$. In the solution above, we calculated a_3 as 15. The formula for a_n can be obtained by first writing down a recurrence relation for it and then solving it. Let S be a set with $2n$ objects. Fix any one object, say x of S . In any pairing, the ‘mate’, say y of x can be chosen in $2n - 1$ ways. Once it is chosen, the problem reduces to pairing off the elements of the set $S - \{x, y\}$ which is a set with $2n - 2$ elements. There are a_{n-1}

pairings of it. So, $\{a_n\}$ satisfies the recurrence relation

$$a_n = (2n - 1)a_{n-1} \quad (1)$$

for $n \geq 2$ with the initial condition $a_1 = 1$. This is very easy to solve by inspection. We start with 1 and thereafter, every time we multiply the earlier term with the next odd number. So

$$a_n = (2n - 1)(2n - 3) \dots \times 5 \times 3 \times 1 \quad (2)$$

By supplying the missing even factors, this can also be written as

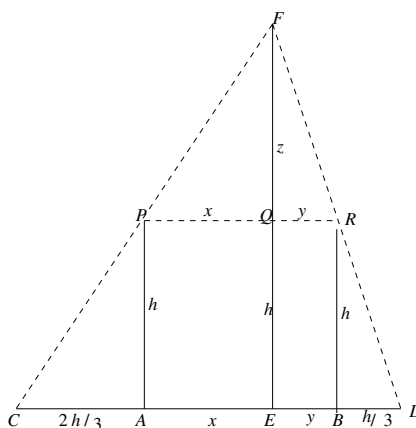
$$\begin{aligned} a_n &= \frac{(2n)!}{(2n)(2n-2)(2n-4) \dots 4 \times 2} \\ &= \frac{(2n)!}{2^n n!} \end{aligned} \quad (3)$$

In particular, $a_3 = \frac{6!}{8 \times 6} = \frac{5!}{8} = \frac{120}{8} = 15$ as we already calculated.

Q.15 Two persons, both of height h are standing at a distance h from each other. The shadow of one person cast by a vertical lamp-post placed between the two persons is double the length of the shadow of the other. If the sum of the lengths of the shadows is h , then the height of the lamp-post is

- (A) $\frac{\sqrt{3}}{2}h$ (B) $2h$ (C) $\left(\frac{1+\sqrt{2}}{2}\right)h$ (D) $\left(\frac{\sqrt{3}+1}{2\sqrt{2}}\right)h$

Answer and Comments: (B). A typical problem about heights and distances. A good diagram is half the solution.



Suppose that the persons are standing at the points A and B and the foot E of the lamp-post EF is at a distance x from A and y from B . Then

$$x + y = h \quad (1)$$

Let P and R be the heads of the persons and let their shadows fall at points C and D on the ground. Then F, P, C are collinear and so are F, R, D . We are given that the lengths of the shadows add to h while one of them is double the length of the other. We take AC to be the larger shadow of length $2h/3$ and BD to be the shorter one of length $h/3$.

We are interested in the length of EF . Normally, we should denote it by a single symbol. However, here we let Q be the point where EF meets PR and let z be the length QF . Then the length of the lamp-post is

$$h + z \quad (2)$$

So we would be through if we can find z in terms of h . The triangles FPQ and PCA are similar. Therefore

$$\frac{z}{x} = \frac{h}{2h/3} = \frac{3}{2} \quad (3)$$

Hence

$$z = \frac{3}{2}x \quad (4)$$

Similarly, the similarity of the triangles FQR and RBD gives

$$z = 3y \quad (5)$$

From (4) and (5), $x = 2y$. Hence from (1), we have $x = 2h/3$ and $y = h/3$. So, from (4) (or (5)) $z = h$. Therefore by (2), the height of the lamp-post is $h + h = 2h$. ■

A simple problem which tests a candidate's ability to visualise the geometric data correctly and to introduce appropriate notations to translate the data. The problem could have been made a little more

demanding by giving the distance between the persons to be different than their common height, and still more demanding by giving the two persons to have different heights. But that would only increase the labour involved without testing any new ability. The paper setters have wisely refrained from such generalisations.

Q.16 Let S be the set of all points z in the complex plane such that

$$\left(1 + \frac{1}{z}\right)^4 = 1.$$

Then, the points of S are

- (A) vertices of a rectangle
- (B) vertices of a right-angled triangle
- (C) vertices of an equilateral triangle
- (D) collinear

Answer and Comments: (D). Every non-zero complex number has n distinct n -th roots. Here we want the fourth roots of 1. They are $\pm 1, \pm i$. Each of these roots will give a separate equation for z and hence (possibly) four different values of z . Let us obtain them one by one.

For the root 1, the equation for z is

$$1 + \frac{1}{z} = 1 \tag{1}$$

which has no solution. The next equation is

$$1 + \frac{1}{z} = -1 \tag{2}$$

whose only solution is $z = -\frac{1}{2}$. For the other two roots, we skip the details and write only the solutions as $z = \frac{1}{i-1}$ and $z = \frac{1}{-i-1}$. Put together,

$$S = \left\{-\frac{1}{2}, \frac{1}{i-1}, \frac{1}{-i-1}\right\} \tag{3}$$

As S has only three elements, option (A) is ruled out. To decide which of the remaining options holds, let us write the second and the third elements of S in the standard form $a + ib$ with a, b real.

$$\frac{1}{i-1} = \frac{-i-1}{(i-1)(-i-1)} = \frac{-1-i}{2} = -\frac{1}{2} - \frac{1}{2}i \quad (4)$$

and similalry,

$$\frac{1}{-i-1} = \frac{i-1}{(-i-1)(i-1)} = \frac{-1+i}{2} = -\frac{1}{2} + \frac{1}{2}i \quad (5)$$

Hence, with these cosmetics,

$$S = \left\{-\frac{1}{2}, -\frac{1}{2} - \frac{1}{2}i, -\frac{1}{2} + \frac{1}{2}i\right\} \quad (6)$$

Clearly all three elements have the same real part, viz. $-\frac{1}{2}$. So all of them lie on the vertical line $x = -\frac{1}{2}$. hence they are collinear. ■

Another simple good problem requiring little more than elementary properties of complex numbers. It is interesting that although the complex number 1 has four distinct fourth roots, the set S has only three elements. This happens because the fourth root 1 of 1 leads to (1) which has no solution in $\mathbb{C}$. Sometimes a hypothetical point ∞ is added to the complex plane to get what is called the **extended complex plane** $\mathbb{C} \cup \{\infty\}$ which is often denoted by $\mathbb{C}^*$. This extra point ∞ is not a complex number. But we allow it to appear in many algebraic expressions by defining

$$z + \infty = z - \infty = \infty \quad (7)$$

for all $z \in \mathbb{C}$,

$$z\infty = \infty \quad (8)$$

for all $z \neq 0$ and

$$\frac{z}{\infty} = 0 \quad (9)$$

for all $z \in \mathbb{C}$. Note that the expressions $0 \times \infty$, $\infty - \infty$ and $\frac{\infty}{\infty}$ are undefined. Also care has to be taken in handling equations involving

∞ . For example, even though $z_1 + \infty = z_2 + \infty$, we may not have $z_1 = z_2$. Despite these limitations, the extended complex plane $\mathbb{C}^*$ is a very convenient device to take care of some anomalies that arise in $\mathbb{C}$. For example, although (1) has no solution in $\mathbb{C}$, it has a solution, ∞ in $\mathbb{C}^*$. With this extra member the solution set S now becomes

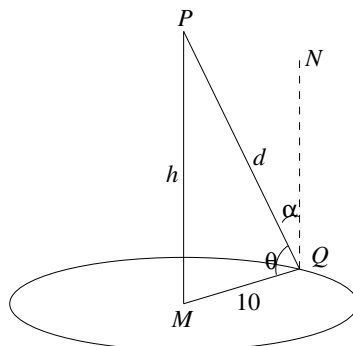
$$S^* = \left\{-\frac{1}{2}, -\frac{1}{2} - \frac{i}{2}, -\frac{1}{2} + \frac{i}{2}, \infty\right\} \quad (10)$$

Collinearity is not lost because ∞ lies on all lines. In fact in $\mathbb{C}^*$, there is no difference between lines and circles. A line is a circle which contains the point ∞ . The points $\pm 1, \pm i$ all lie on the unit circle $|z| = 1$. This circle is the image of the line $x = -\frac{1}{2}$ under the transformation which takes z to $1 + \frac{1}{z} = \frac{z+1}{z}$. This transformation belongs to an important class of transformations of $\mathbb{C}^*$ to $\mathbb{C}^*$ called the **fractional linear transformations** because they are of the form $\frac{az+b}{cz+d}$ i.e. ratios of two linear transformations. We could go on in this wonderland of $\mathbb{C}^*$. But that is not relevant to the present problem. We mentioned it because those who study complex numbers in some depth are sure to encounter $\mathbb{C}^*$.

- Q.17 A circular lawn of diameter 20 meters on a horizontal plane is to be illuminated by a light source placed vertically above the centre of the lawn. It is known that the illuminance at a point P on the lawn is given by the formula $I = \frac{C \sin \theta}{d^2}$ for some constant C , where d is the distance of P from the light source and θ is the angle between the line joining the centre of the lawn to P and the line joining the light source to P . Then the maximum possible illuminance at a point on the circumference of the lawn is

(A) $\frac{C}{75\sqrt{3}}$ (B) $\frac{C}{100\sqrt{3}}$ (C) $\frac{C}{150\sqrt{3}}$ (D) $\frac{C}{250\sqrt{3}}$

Answer and Comments: (C). As usual, we begin with a diagram.



If a light source is located at a point P , then the illuminance of it at a point Q is directly proportional to the intensity of the source (which is denoted by the constant C here and has little role in the problem) and inversely proportional to the square of the distance between P and Q (which is denoted here by d). It is also proportional directly to the cosine of the angle (shown by α) the line PQ makes with the normal (shown by QN) to the surface on which Q lies. As a result, the smaller this angle the more is the illuminance. (The same is true for sources that radiate other forms of energy such as heat. A common application of this is in the design of solar heaters where the panel that captures the heat from the sun rays is kept sloping so that it would capture more energy during the winter months.)

In the present problem, this surface is the horizontal plane and therefore the normal to it at any point Q is simply the vertical line through Q . Instead of expressing the law of illuminance in terms of α , it is given in terms of the complementary angle θ .

It should have been made clear in the data that the vertical height (denoted by h in the figure) of the light source is variable. Normally, we deal with a light source which is fixed. In the present problem, if h is very small (i.e. if the light source is closer to the ground), then d and hence d^2 will also be small. But we may not get maximum illuminance because the angle θ and hence its sine will also be small. The higher the height h , the greater is the distance d but so is $\sin \theta$. It would have been less confusing to pose the problem as the problem of determining the height for which every point on the circumference will have maximum illuminance. Already the text of the problem is long. Also to understand the data, a candidate has to draw a diagram.

If, further, the candidate has to figure out exactly the purpose of the problem, it is unfair when the time allotted is meagre (four minutes).

Fortunately, in the present problem, once these hurdles are crossed, the rest of the work is simple. We express the illuminance as a function of θ and maximise it. Clearly

$$d = 10 \sec \theta \quad (1)$$

and so the illuminance, as a function of θ is

$$I = I(\theta) = \frac{C \sin \theta}{100 \sec^2 \theta}$$

Simplifying,

$$\begin{aligned} I(\theta) &= \frac{C}{100} \sin \theta \cos^2 \theta \\ &= \frac{C}{100} (\sin \theta - \sin^3 \theta) \end{aligned} \quad (2)$$

So the problem reduces to maximising the function $\sin \theta - \sin^3 \theta$ as θ varies from 0 to $\frac{\pi}{2}$. As we are not interested in the value of θ at which the maximum occurs, we might as well put $x = \sin \theta$ and maximise the function $f(x) = x - x^3$ for $x \in [0, 1]$. The only critical point of f in $[0, 1]$ is given by $1 = 3x^2$ whence $x = \frac{1}{\sqrt{3}}$. By a direct substitution,

$$f\left(\frac{1}{\sqrt{3}}\right) = \frac{1}{\sqrt{3}} - \frac{1}{3\sqrt{3}} = \frac{2}{3\sqrt{3}}. \text{ Hence the maximum illuminance is } \frac{2C}{300\sqrt{3}} = \frac{C}{150\sqrt{3}}. \blacksquare$$

A reasonable problem on maximisation, except for the longish and slightly confusing wording. The problem would have a more natural appeal if it was posed as a problem of determining the location of the light source (directly above the centre of the lawn) that will maximise the illuminance at all points on the circumference of the lawn. Nor would it entail much extra work. After determining $\sin \theta$ as $\frac{1}{\sqrt{3}}$ for maximisation, $h = 10 \tan \theta = \frac{10}{\sqrt{2}}$ meters.

Q.18 Let f and g be two real-valued, continuous functions defined on the closed interval $[a, b]$, such that $f(a) < g(a)$ and $f(b) > g(b)$. Then the area enclosed between the graphs of the two functions and the lines $x = a$ and $x = b$ is always given by

- (A) $\int_a^b |f(x) - g(x)| dx$ (B) $|\int_a^b f(x) - g(x) dx|$
 (C) $\left| \int_a^b |f(x)| - |g(x)| dx \right|$ (D) $\int_a^b ||f(x)| - |g(x)|| dx$

Answer and Comments: (A). The classical geometric interpretation of a definite integral $\int_a^b f(x) dx$ is that it is the ‘area $A(R)$ of the region, say R , below the graph of $f(x)$ ’. Here it is understood that R is bounded on the sides by the lines $x = a$ and $x = b$. More importantly, it is understood that the lower boundary of the R is the x -axis. This presupposes that $f(x)$ is non-negative throughout the interval $[a, b]$.

If $f(x) \leq 0$ for all $x \in [a, b]$, then the integral $\int_a^b f(x) dx$ is negative. In such a case, in order that it equals the area, we have to take the area below the x -axis as negative. But as a geometric entity the area is non-negative. So, in such a case, the area is the integral $\int_a^b -f(x) dx$ and we call it the area ‘below’ rather than ‘above’ the graph $y = f(x)$.

Things get more complicated when $f(x)$ is non-negative on some parts of $[a, b]$ and negative on the others. Then the graph $y = f(x)$ lies above the x -axis in some parts and below it in others and the integral of f has little geometric significance. In fact, the integral $\int_a^b f(x) dx$ may even vanish (as it always does when $f(x)$ is an odd function and $b = -a$). However, if we take $\int_a^b |f(x)| dx$, then it measure the area ‘between’ the graph $y = f(x)$ and the x -axis. Of course, to evaluate this area, we need to split $[a, b]$ into several subintervals, over each of which $f(x)$ maintains its sign. This can be done by identifying the zeros of $f(x)$ in $[a, b]$. If they are $c_1, c_2, \dots, c_r$ in ascending order, we set $c_0 = a, c_{r+1} = b$. Then on each subinterval $[c_{i-1}, c_i]$, either $|f| = f$ or $|f| = -f$.

Note that the x -axis can be thought of as the graph of the identically

zero function $g(x)$. More generally, when $f(x)$ and $g(x)$ are any two functions, in some parts of $[a, b]$ we may have $f(x) \geq g(x)$ while in the rest we have $f(x) \leq g(x)$. But no matter which possibility holds where, $\int_a^b |f(x) - g(x)| dx$ is the area between the graphs of $f(x)$ and $g(x)$. So (A) is correct. Once again, this trick merely gives a succinct formula for the area. When it comes to actually calculate the area between the graphs $y = f(x)$ and $y = g(x)$, there is no short cut to splitting the interval $[a, b]$ into subintervals, in each of which $f(x)$ lies either entirely above or entirely below $g(x)$. For example, to calculate the area between the graphs of $y = \sin x$ and $y = \cos x$ for $0 \leq x \leq \pi$, we need to split it into $[0, \pi/4]$ and $[\pi/4, \pi]$. ■

The present problem tests whether the candidates have been taught this care. The hypothesis $f(a) < g(a)$ and $f(b) > g(b)$ is really redundant as the question asks to pick up the choice which is *always* correct. The hypothesis merely draws attention to the fact in this case it is mandatory to take the integrand as $|f(x) - g(x)|$.

Q.19 Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as follows:

$$f(x) = \begin{cases} (x-1) \min\{x, x^2\} & \text{if } x \geq 0 \\ x \min\{x, \frac{1}{x}\} & \text{if } x < 0 \end{cases}$$

Then f is

- (A) differentiable everywhere
- (B) *not* differentiable at exactly one point
- (C) *not* differentiable at exactly two points
- (D) *not* differentiable at exactly three points.

Answer and Comments: (C). Yet another problem where a function f has different definitions at different points of the domain. While testing the differentiability of such a function, one has to first identify these turning points where a change of formula occurs and then at each of these possible trouble makers, check differentiability from the left and from the right. Usually, separate arguments are needed at each such point and on each of the two sides.

In the present problem, one obvious point where the function f changes its definition is, of course, 0. It is given right in the statement of the problem. But usually there are some hidden points too. For $x \geq 0$, x and x^2 are equal at 1. But in the interval $(0, 1)$, x^2 is smaller than x while for $x > 1$, x is smaller than x^2 . So, for $x \geq 0$, the definition of $f(x)$ becomes

$$f(x) = \begin{cases} x^3 - x^2 & \text{if } 0 \leq x \leq 1 \\ x^2 - x & \text{if } x > 1 \end{cases} \quad (1)$$

We now have all the information we need to test differentiability of f at 1 and also its right differentiability at 0. Both the functions $x^3 - x^2$ and $x^2 - x$ are continuously differentiable everywhere and so their derivatives at any point c can be safely obtained by considering their derivatives at x and letting $x \rightarrow c$. (This reasoning fails for functions whose derivatives are not continuous. For example, if $g(x) = x^2 \sin(1/x)$ for $x \neq 0$ and $g(0) = 0$, then $g'(0)$ can be shown to exist from first principles. But it cannot be obtained by letting $x \rightarrow 0$ in $2x \sin(1/x) - \cos(1/x^2)$ because the second term does not tend to any limit.) So, we have

$$f'_-(1) = \lim_{x \rightarrow 1^-} 3x^2 - 2x = 1 \quad (2)$$

$$f'_+(1) = \lim_{x \rightarrow 1^+} 2x - 1 = 1 \quad (3)$$

$$\text{and } f'_+(0) = \lim_{x \rightarrow 0^+} 3x^2 - 2x = 0 \quad (4)$$

From (2) and (3), we see that f is differentiable at 1. Also since $f(0) = 0$, $f(x)$ is continuous from the right at 0 and (4) gives us one sided information about its differentiability at 0. To see what happens on the other side of the border, i.e. for $x < 0$. We first need the analogue of (1). At $x = -1$, x and $\frac{1}{x}$ equal each other. But if $-1 < x < 0$, then $\frac{1}{x} < x$ while for $x < -1$, $x < \frac{1}{x}$. (To see such things more easily, give some particular values to x , e.g. $x = -\frac{1}{2}$ in the first case and $x = -10$ in the second.) So analogously to (1) we have

$$f(x) = \begin{cases} x^2 & \text{if } x < -1 \\ 1 & \text{if } -1 \leq x < 0 \end{cases} \quad (5)$$

Clearly, f is continuous at -1 . But at $x = 0$, $\lim_{x \rightarrow 0^-} f(x) = 1$ while $f(0) = 0$. So f is not continuous at 0 and hence automatically not differentiable at 0 either.

It remains to check differentiability of f at -1 . Analogously to (2) and (3) we have

$$f'_-(-1) = \lim_{x \rightarrow -1^-} 2x = -2 \quad (6)$$

$$\text{and } f'_+(-1) = \lim_{x \rightarrow -1^+} 0 = 0 \quad (7)$$

From (6) and (7), f is not differentiable at -1 .

All put together f is differentiable everywhere except at the two points -1 and 0 . ■

The problem is simple once we identify the intervals on which f is given by a single formula, viz., $(-\infty, -1)$, $[-1, 0)$, $[0, 1]$ and $(1, \infty)$. The only catch is that those who miss that f is discontinuous at 0 may falsely conclude its differentiability at 0 from (4) and the fact that $\lim_{x \rightarrow 0^-} f'(x) = \lim_{x \rightarrow 0^-} 0 = 0$. When a function $f(x)$ is discontinuous at a point c , and $\lim_{x \rightarrow c^-} f'(x)$ exists and equals L (say), it *does not* follow that $f'_-(c) = L$. This would be the case if f is known to be continuous at c . Even then, the proof is not trivial. It needs Lagrange's Mean Value Theorem applied to f on an interval of the form $[c - h, c]$ for $h > 0$.

Q.20 Let $f : [0, 2] \rightarrow \mathbb{R}$ be a continuous function such that

$$\frac{1}{2} \int_0^2 f(x) \, dx < f(2).$$

Then which of the following statements must be true?

- (A) f must be strictly increasing.
- (B) f must attain a maximum value at $x = 2$.
- (C) f cannot have a minimum at $x = 2$.
- (D) None of the above.

Answer and Comments: (C). The L.H.S. of the given inequality is the average value, say A , of f over the interval $[0, 2]$, while the R.H.S. is the value of f at a particular point c in the domain. (Here c happens to be the right end point of the interval $[0, 2]$. But that makes no difference.)

When the average is less than a particular functional value $f(c)$, it does not follow that the function is less than $f(c)$ throughout. This is sheer common sense. If the average income of some community is, say Rs.10,000.00 and some particular person x has an income of Rs. 50,000 it does not follow that he is the richest member of the community. The community may very well have some persons richer than x , but many other persons poorer than x can pull the average down. All that can be said definitely is that such a person cannot be the poorest person, for if he were then everybody's income will be at least Rs. 50,000.00 and hence so will be the average income for the community.

The present problem requires the continuous analogue of the reasoning in this example. Suppose (C) fails, that is f has a minimum at 2. Then

$$f(x) - f(2) \geq 0 \quad (1)$$

for all $x \in [0, 2]$. So

$$\int_0^2 f(x) - f(2) \, dx \geq 0 \quad (2)$$

whence

$$\int_0^2 f(x) \, dx \geq \int_0^2 f(2) \, dx = 2f(2) \quad (3)$$

Dividing both the sides by the positive number 2, we get

$$\frac{1}{2} \int_0^2 f(x) \, dx \geq f(2) \quad (4)$$

contradicting the hypothesis. So (C) holds. ■

An absolutely simple problem once the idea of the average value strikes.

Q.21 In a triangle ABC , $3 \sin A + 4 \cos B = 6$ and $4 \sin B + 3 \cos A = 1$ hold. Then the angle C equals

- (A) 30° (B) 60° (C) 120° (D) 150°

Answer and Comments: (A). The three angles A, B, C always add to 180 degrees. So, there are only two independent angles and in general two equations in them will determine them. In the present problem, these equations are given to be

$$3 \sin A + 4 \cos B = 6 \quad (1)$$

$$\text{and } 4 \sin B + 3 \cos A = 1 \quad (2)$$

We can eliminate either one of A and B to get an equation for the other angle. For example, writing $3 \sin A = (6 - 4 \cos B)$ and $3 \cos A = (1 - 4 \sin B)$, squaring and adding gives

$$9 = (6 - 4 \cos B)^2 + (1 - 4 \sin B)^2 \quad (3)$$

which simplifies to

$$2 \sin B + 12 \cos B = 11 \quad (4)$$

Similarly by squaring $4 \cos B = 6 - 3 \sin A$ and $4 \sin B = 1 - 3 \cos A$ and adding we get

$$16 = (6 - 3 \sin A)^2 + (1 - 3 \cos A)^2 \quad (5)$$

which reduces to

$$6 \sin A + \cos A = 5 \quad (6)$$

We can divide both the sides of (4) by $\sqrt{148}$ and rewrite it as

$$\frac{1}{\sqrt{37}} \sin B + \frac{6}{\sqrt{37}} \cos B = \frac{11}{\sqrt{148}} \quad (7)$$

and further as

$$\cos(B - \alpha) = \frac{11}{\sqrt{148}} \quad (8)$$

where α is the unique angle for which $\cos \alpha = \frac{6}{\sqrt{37}}$ and $\sin \alpha = \frac{1}{\sqrt{37}}$. Then we would get

$$B = \cos^{-1}\left(\frac{11}{\sqrt{148}}\right) + \alpha \quad (9)$$

Similarly, from (6) we can determine A . Having found A and B , C will come out as $\pi - A - B$. But it will require a lot of calculation to determine C in degrees. So we look for a simpler solution, keeping in mind that

$$\sin C = \sin A \cos B + \cos A \sin B \quad (10)$$

always holds in any triangle because $\sin C = \sin(\pi - (A + B)) = \sin(A + B)$. We simply multiply (1) by $\cos B$ and (2) by $\sin B$ and add to get

$$3(\sin A \cos B + \cos A \sin B) = 6 \cos B + \sin B - 4 \quad (11)$$

Combining this with (10) and (4), we get

$$3 \sin C = \frac{11}{2} - 4 = \frac{3}{2} \quad (12)$$

which gives

$$\sin C = \frac{1}{2} \quad (13)$$

So either $C = 30^\circ$ or $C = 150^\circ$. If the second possibility holds then $A + B = 30^\circ$. That means both the angles A and B would be smaller than 30° . Hence their cosines will be greater than $\frac{\sqrt{3}}{2}$. But that will contradict (2) because $\sin B > 0$ and $3 \cos A > \frac{3\sqrt{3}}{2} > 1$. So we have $C = 30^\circ$. ■

A good computational problem which tests the skill to choose the right method. Note that although we abandoned (8) which followed from (4), we did use (4). Instead of (4) we could have used (6) which is slightly simpler. It is usually difficult to tell right at the start which approach would be simpler and do only the work needed for it. One has to be prepared for the possibility that some of the work will turn out to be superfluous later. But that is the name of the game. If

somebody is really bent on using both (4) and (6), a salvaging feature is that from (6) we get $\cos(A - \beta) = \frac{5}{\sqrt{37}}$ where β is the complementary angle of the angle α in (9). So, adding, $A + B$ would come out to be $\frac{\pi}{2} + \cos^{-1}(\frac{11}{\sqrt{148}}) + \cos^{-1}(\frac{6}{\sqrt{37}})$. This can indeed be simplified to 150° .

It is important to note that such a succinct answer as 30° was possible only because the numerical data permitted it. If, instead of (1) and (2), we had a more general system of equations, viz.

$$a_1 \sin A + b_1 \cos B = c_1 \quad (14)$$

$$\text{and } a_2 \sin B + b_2 \cos A = c_2 \quad (15)$$

our first method would still work and will give B and A and finally C . But the answers would be horrendous. The paper setters deserve to be commended for giving the numerical data which makes the solution manageable.

Q.22 Let $\theta = \frac{2\pi}{7}$ and consider the following matrix

$$A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

If A^n means $A \times A \times \dots \times A$ (n times), then A^{100} is

$$\begin{array}{ll} \text{(A)} \begin{pmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{pmatrix} & \text{(B)} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \\ \text{(C)} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} & \text{(D)} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \end{array}$$

Answer and Comments: (A). As the matrix A depends on θ , denote it by A_θ . We can similarly define the matrices A_α , A_β etc. for any real numbers α, β etc. A direct calculation gives

$$\begin{aligned} A_\alpha A_\beta &= \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} \cos \beta & -\sin \beta \\ \sin \beta & \cos \beta \end{pmatrix} \\ &= \begin{pmatrix} \cos \alpha \cos \beta - \sin \alpha \sin \beta & -\sin \alpha \cos \beta - \cos \alpha \sin \beta \\ \sin \alpha \cos \beta + \cos \alpha \sin \beta & \cos \alpha \cos \beta - \sin \alpha \sin \beta \end{pmatrix} \\ &= \begin{pmatrix} \cos(\alpha + \beta) & -\sin(\alpha + \beta) \\ \sin(\alpha + \beta) & \cos(\alpha + \beta) \end{pmatrix} \\ &= A_{\alpha+\beta} \end{aligned} \quad (1)$$

for all α, β . In particular, taking $\alpha = \beta = \theta$, we have

$$A_\theta^2 = A_{2\theta} \quad (2)$$

and more generally, by induction on n ,

$$A_\theta^n = A_{n\theta} \quad (3)$$

In particular, for $\theta = 2\pi/7$,

$$A^{100} = A_{2\pi/7}^{100} = \begin{pmatrix} \cos(200\pi/7) & -\sin(200\pi/7) \\ \sin(200\pi/7) & \cos(200\pi/7) \end{pmatrix} \quad (4)$$

From $\frac{200\pi}{7} = 28\pi + \frac{4\pi}{7}$ and the periodicity of the trigonometric functions, A^{100} is the same as $\begin{pmatrix} \cos(4\pi/7) & -\sin(4\pi/7) \\ \sin(4\pi/7) & \cos(4\pi/7) \end{pmatrix}$. So (A) is correct. ■

Property (1) is fairly well-known. It is pivotal in deriving (3) on which the solution to the present problem is based. An alternate derivation of (3) is possible using complex numbers. Every matrix of the form $\begin{pmatrix} x & -y \\ y & x \end{pmatrix}$ where x, y are real numbers, corresponds to the complex number $z = x + iy$. This association preserves complex addition and multiplication. That is, if $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$ are any two complex numbers, and we denote their corresponding matrices by A_{z_1} and A_{z_2} respectively, then a direct calculation gives

$$A_{z_1+z_2} = A_{z_1} + A_{z_2} \quad (5)$$

$$\text{and } A_{z_1 z_2} = A_{z_1} A_{z_2} \quad (6)$$

Note that the matrix A_θ in the statement of the problem is the same as the matrix $A_{\cos\theta + i\sin\theta}$ associated with the complex number $\cos\theta + i\sin\theta$. Therefore, by repeated applications of (6), $(A_\theta)^n$ corresponds to the complex number $(\cos\theta + i\sin\theta)^n$. By DeMoivre's rule,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta \quad (7)$$

Therefore, (3) holds for every positive integer n . (Some persons denote the number $\cos\theta + i\sin\theta$ by $e^{i\theta}$. Then a quick proof of (7) comes from $(e^{i\theta})^n = e^{in\theta}$. But unless we have independently defined and proved

the basic properties of the complex exponential function, this proof is nothing more than a notational gimmick.)

Yet another interpretation of the matrix A_θ is provided by the concept of a rotation. Let $T_\theta : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ denote an anticlockwise rotation of the plane around the origin O , through an angle θ . Clearly $T_\theta(O) = O$. For any other point $P = (x, y)$ in $\mathbb{R}^2$, $T_\theta(P)$ can be calculated as follows. Let r and α be the polar coordinates of P . Then $r = \sqrt{x^2 + y^2}$ and α is the angle OP makes with the positive x -axis. The cartesian coordinates x, y can be obtained from r and α by

$$x = r \cos \alpha, \quad y = r \sin \alpha \quad (8)$$

Now let $Q = T_\theta(P)$. Then $OQ = OP = r$. But OQ makes an angle $\alpha + \theta$ with the positive x -axis. Hence the polar coordinates of Q are r and $\theta + \alpha$. Therefore, if the cartesian coordinates of Q are x', y' , then

$$\begin{aligned} x' &= r \cos(\theta + \alpha) \\ &= r(\cos \theta \cos \alpha - \sin \theta \sin \alpha) \\ &= \cos \theta x - \sin \theta y \\ &= (\cos \theta \quad -\sin \theta) \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned} \quad (9)$$

where the R.H.S. of (9) is the product of a 1×2 matrix and a 2×1 matrix. By an entirely analogous calculation, we get

$$y' = (\sin \theta \quad \cos \theta) \begin{pmatrix} x \\ y \end{pmatrix} \quad (10)$$

(9) and (10) can be combined together to give

$$\begin{aligned} \begin{pmatrix} x' \\ y' \end{pmatrix} &= \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} \\ &= A_\theta \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned} \quad (11)$$

So, if we denote the points P and Q of $\mathbb{R}^2$ by the column vectors $\begin{pmatrix} x \\ y \end{pmatrix}$ and $\begin{pmatrix} x' \\ y' \end{pmatrix}$ respectively, rather than by the ordered pairs (x, y)

and (x', y') which is more usual, we see that the effect of applying the rotation T_θ on P is the same as premultiplying the column vector $\begin{pmatrix} x \\ y \end{pmatrix}$ by the matrix A_θ .

Now consider two rotations T_α and T_β . It is clear that if we first apply T_β and then T_α , the net result is the rotation $T_{\alpha+\beta}$. In other words

$$T_{\alpha+\beta} = T_\alpha \circ T_\beta \quad (12)$$

Given any point $P = \begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$, let $Q = T_\beta(P)$ and $R = T_\alpha(Q)$.

Denote Q and R by the column vectors $\begin{pmatrix} x' \\ y' \end{pmatrix}$ and $\begin{pmatrix} x'' \\ y'' \end{pmatrix}$ respectively.

We apply (11) repeatedly, first with $\theta = \alpha$, then with $\theta = \beta$ to get

$$\begin{aligned} \begin{pmatrix} x'' \\ y'' \end{pmatrix} &= A_\alpha \begin{pmatrix} x' \\ y' \end{pmatrix} \\ &= A_\alpha A_\beta \begin{pmatrix} x \\ y \end{pmatrix} \end{aligned} \quad (13)$$

But on the other hand, applying (11) with $\theta = \alpha + \beta$,

$$\begin{pmatrix} x'' \\ y'' \end{pmatrix} = A_{\alpha+\beta} \begin{pmatrix} x \\ y \end{pmatrix} \quad (14)$$

Combining (13) and (14),

$$A_{\alpha+\beta} \begin{pmatrix} x \\ y \end{pmatrix} = A_\alpha A_\beta \begin{pmatrix} x \\ y \end{pmatrix} \quad (15)$$

for all $x, y \in \mathbb{R}^2$. From this we cannot hastily conclude that $A_{\alpha+\beta} = A_\alpha A_\beta$, because for matrices it may very well happen that $AC = BC$ and still $A \neq B$, even when the matrix C is not the zero matrix. But in the present case, we are given that (15) holds for *all* x, y . For any 2×2 matrix C , it is easy to check by a direct computation that $C \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is simply the first column of C . Hence applying (15) with $x = 1, y = 0$ we

get that the first column of $A_{\alpha+\beta}$ is the same as that of $A_\alpha A_\beta$. Similarly by applying (15) with $x = 0, y = 1$ we get that their second columns are the same. We are now justified in saying that we have an alternate proof of (1).

Of course, a proof based on a direct computation is much shorter. But interpreting matrices as transformations and their product as the composite of transformations opens a whole new branch of mathematics called **linear algebra**. The theory of determinants or that of systems of linear equations properly belongs to this branch of mathematics.

Q.23 Consider all the permutations of the twenty six English letters that start with z . In how many of these permutations the number of letters between z and y is less than those between y and x ?

- (A) $6 \times 23!$ (B) $6 \times 24!$ (C) $156 \times 23!$ (D) $156 \times 24!$

Answer and Comments: (B). The dependence on the English alphabet is superficial. The knowledge that the last three letters of the alphabet are x, y, z is nowhere needed in the solution. The question could as well have been posed for permutations of any set S with 26 elements which satisfy the given condition for three specified elements of S . Such an unwarranted involvement of something extraneous is, in fact, likely to cause some confusion (which sometimes leads to a controversy) because candidates who have opted for some other language as their medium of instruction and are diffident about their knowledge of English may get dissuaded by (incorrectly) thinking that the problem is beyond their scope.

It is argued, and with some justification, that such settings impart a certain liveliness to a problem. If every problem in mathematics is to be posed only after stripping it of these real life links, mathematics would appear more dull and would be feared more than it is today. But in an examination, care has to be taken to ensure that these links do not add to a candidate's burden. For example, there is nothing wrong in asking a question about dealing the cards of an ordinary deck. But it should not require a candidate to know the rules of a particular card game.

And, if the paper setters are so much in love with the English alpha-

bet, one certainly wishes that they used the correct English grammar! The problem is in the form of a direct question. The verb then has to be split and the auxiliary part of it must come before the subject. Thus the word ‘is’ ought to have come before the phrase ‘the number’.

Anyway coming to the mathematical essence of the problem, suppose the gap between z and y is m letters. Here m can be 0, if there is no gap, i.e. y comes immediately after z . Now since the gap between y and x is more than m , x cannot come between z and y . So it has to come after x and after a gap of at least $m + 1$ letters.

We classify the permutations according to the value of m . When $m = 0$, the permutation begins with

$$zy \times$$

where the place $\times$ is to be filled by any of the 23 letters (i.e. those other than x, y, z). But after that the remaining 23 places can be filled freely by the remaining 23 letters (of which x is one). So the number of permutations where $m = 0$ is

$$23 \times (23 \times 22 \times 21 \times \dots \times 2 \times 1) = 23 \times 23! \quad (1)$$

Let us now suppose that $m = 1$. Then the permutation must begin with

$$z \times y \times \times \quad (2)$$

where the three places marked with an $\times$ are to be filled by any of the 23 symbols (letters other than x, y, z). This can be done in $23 \times 22 \times 21$ ways. We now have 21 letters left (of which x is one) and they can occupy these remaining 21 places in any order. So the number of permutations of the desired type with $m = 1$ is

$$23 \times 22 \times 21 \times (21!) = 21 \times 23! \quad (3)$$

The general pattern should be clear from (1) and (3). We expect that for $m = 2$ the number of permutations will be

$$19 \times 23! \quad (4)$$

which can be verified by observing that every such permutation must begin with

$$z \times \times y \times \times \times \quad (5)$$

Thus every time we are getting $23!$ times an odd number. These odd numbers start from 23, 21, 19, ... So they will end with 3 and finally 1. (We can determine the m for which the coefficient of $23!$ would be 1. But that is only an additional labour.) So the answer to the problem is

$$23! \times (23 + 21 + 19 + \dots + 5 + 3 + 1) \quad (6)$$

The expression in the parentheses is the sum of an A.P. of length 12. There is, of course, a formula for it. But the best way is to add them the way the great mathematician **Gauss** did when he was 7 years old. We add together the two end figures 23 and 1 to get 24. Then we add the ends of the remaining, i.e. 21 and 3 to still get 24. This will go on till we add the middle two numbers 13 and 11 which also gives 24. So the total count is

$$6 \times 24 \times 23! = 6 \times 24! \quad (7)$$

which tallies with (B). ■

This is an excellent problem which tests the ability to experiment, see some pattern and come up with a correct guess. The mathematical excellence of the problem outweighs the minor slip of grammar mentioned earlier.

Q.24 Let $P = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and $Q = \left(-\frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}\right)$ be two vertices of a regular polygon having 12 sides such that PQ is a diameter of the circle circumscribing the polygon. Which of the following points is not a vertex of this polygon ?

- | | |
|---|---|
| (A) $\left(\frac{\sqrt{3}-1}{2\sqrt{2}}, \frac{\sqrt{3}+1}{2\sqrt{2}}\right)$ | (B) $\left(\frac{\sqrt{3}+1}{2\sqrt{2}}, \frac{\sqrt{3}-1}{2\sqrt{2}}\right)$ |
| (C) $\left(\frac{\sqrt{3}+1}{2\sqrt{2}}, \frac{1-\sqrt{3}}{2\sqrt{2}}\right)$ | (D) $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ |

Answer and Comments: (D). Both P and Q are at a distance 1 from the origin O which must also be the centre of the polygon since P

and Q are given to be diametrically opposite to each other. Hence all vertices of it are at a distance 1 from the origin. By a direct calculation, this is the case for the points in all the four options. So, we cannot get the odd man out using this preliminary criterion. We need to dig deeper.

The angular distance between any two consecutive vertices of the polygon is $\frac{2\pi}{12} = \frac{\pi}{6}$. The ray OP makes an angle $\frac{\pi}{4}$ with the x -axis. So for every other vertex, say V of the polygon the angle OV makes with the x -axis will be of the form $\frac{\pi}{4} + \frac{k\pi}{6}$ where the integer k will run from 1 to 11. (For $k = 6$ we get the angle $\pi + \frac{\pi}{4}$ and the corresponding vertex is Q as expected.)

So, to nab the culprit, for each of the given four options, say V , we determine the angle which OV makes with the x -axis. Here we are lucky. The number in the option (D) is more popularly denoted by ω . It is an imaginary cube root of 1. It can be written in the form $(\cos(2\pi/3), \sin(2\pi/3))$. So if V is the vertex in (D), then the angle OV is $\frac{2\pi}{3}$. If this were to be a vertex of the polygon, then $\frac{2\pi}{3} - \frac{\pi}{4}$ will have to be an integral multiple of $\frac{\pi}{6}$. But by a direct calculation, $\frac{2\pi}{3} - \frac{\pi}{4} = \frac{5\pi}{12}$ which is not an integral multiple of $\frac{\pi}{6}$. So (D) is correct. ■

A good, but somewhat unusual problem. An exhaustive trial of all the four options, one by one, until we identify the culprit would have been very time consuming. The paper setters have been helpful to those who recognise one of the options as a more familiar complex number than the others and luckily that being the correct option.

Q.25 Let a, b, c be real numbers such that $a = a^2 + b^2 + c^2$. What is the smallest possible value of b ?

- (A) 0 (B) -1 (C) $-\frac{1}{4}$ (D) $-\frac{1}{2}$

Answer and Comments: (D). Let us write the given equation as

$$a - a^2 = b^2 + c^2 \quad (1)$$

The R.H.S. is non-negative. The L.H.S. is a quadratic in a with a negative leading coefficient. So its value is non-negative only when a lies between the two roots 0 and 1. Also the maximum value of the L.H.S. occurs at $a = \frac{1}{2}$ and this maximum is $\frac{1}{4}$. So this is the maximum $b^2 + c^2$ can be. Subject to this, we have to minimise b . This will be the case when b^2 is the maximum and b is the negative square root of this maximum. For b^2 to be maximum, the L.H.S. should be maximum and c^2 should be minimum. So, we take $a = \frac{1}{2}$ and $c = 0$. Then $b^2 = \frac{1}{4}$. Therefore

$$-\frac{1}{2} \leq b \leq \frac{1}{2} \quad (2)$$

Hence the minimum of b is $-\frac{1}{2}$. For any smaller value of b , b^2 will exceed $\frac{1}{4}$ and then (1) cannot hold for any a and c . ■

A simple problem on minimisation but does not fall into any standard type such as the calculus method or the A.M.-G.M. inequality. It is a good test of the ability to think fresh rather than be guided by a large number of problems of the familiar type.

Q.26 Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$ defined as

$$f(x) = \begin{cases} \frac{x}{e^x - 1} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$$

Then which of the following statements is correct?

- (A) f is not continuous at $x = 0$.
- (B) f is continuous but not differentiable at $x = 0$.
- (C) f is differentiable at $x = 0$ and $f'(0) = -\frac{1}{2}$.
- (D) f is differentiable at $x = 0$ and $f'(0) = \frac{1}{2}$.

Answer and Comments: (C). To test the continuity of f at 0, we need to find $\lim_{x \rightarrow 0} \frac{x}{e^x - 1}$. It is much easier to find the limit of the recipro-

cal, i.e. $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$ because this limit is, by very definition, the derivative of the function e^x at 0. It exists and equals 1. So $\lim_{x \rightarrow 0} f(x) = \frac{1}{1} = 1$. Hence f is continuous at 0.

For differentiability too, we can consider the reciprocal of $f(x)$. But unlike limits, the derivative of a function is not so easily related to that of its reciprocal. So we proceed directly. We have to consider the limit, say L , defined by

$$\begin{aligned} L &= \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} \\ &= \lim_{x \rightarrow 0} \frac{\frac{x}{e^x - 1} - 1}{x} \\ &= \lim_{x \rightarrow 0} \frac{x - e^x + 1}{x(e^x - 1)} \end{aligned} \quad (1)$$

This is a limit of the indeterminate form $\frac{0}{0}$. So we try l'Hôpital's rule to evaluate it. Taking the ratio of the derivatives of the numerator and the denominator, the limit L will be

$$L = \lim_{x \rightarrow 0} \frac{1 - e^x}{xe^x + e^x - 1} \quad (2)$$

provided this limit exists. But this is again of the $\frac{0}{0}$ form. So, we invoke l'Hôpital's rule again to get

$$L = \lim_{x \rightarrow 0} \frac{-e^x}{xe^x + 2e^x} \quad (3)$$

provided, again, that this limit exists. But this time we have no trouble. As $x \rightarrow 0$, the numerator tends to -1 while the denominator tends to 2. So $L = -\frac{1}{2}$. Therefore $f'(0)$ exists and equals $-\frac{1}{2}$. ■

A good problem requiring two applications of the l'Hôpital's rule. It is to be expected that the rule will be needed twice, because in the ratio (1) the denominator has a zero of order 2 at 0 since both the factors x and $e^x - 1$ vanish.

The problem can also be solved by advanced methods, which are often used at the Junior College level without justification. For example, we can take the Taylor series of e^x at 0 (also called MacLaurin series) viz.

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} + \dots \quad (4)$$

If we substitute this into (1), the numerator becomes

$$-\frac{x^2}{2!} - \frac{x^3}{3!} - \dots - \frac{x^n}{n!} - \dots \quad (5)$$

while the denominator becomes

$$x^2 + \frac{x^3}{2!} + \frac{x^4}{3!} + \dots + \frac{x^{n+1}}{n!} + \dots \quad (6)$$

In both (5) and (6), the dominating power near 0 is x^2 . So the ratio of (5) by (6) tends to the ratio of the coefficients of this power, i.e. $-\frac{1}{2}$.

What is involved in this argument is not only the power series expansion of e^x , but also the operation of equating the limit of a sum of infinitely many terms with the sum of the limits of the individual terms. Such arguments need justifications. So, it would have been better to ask this question in Paper 2 which would have shown what type of justification the candidate has given.

Q.27 Let the function $f : [0, 1] \rightarrow \mathbb{R}$ be defined as

$$f(x) = \max \left\{ \frac{|x - y|}{x + y + 1} : 0 \leq y \leq 1 \right\} \text{ for } 0 \leq x \leq 1.$$

Then which of the following statements is correct?

- (A) f is strictly increasing on $[0, \frac{1}{2}]$ and strictly decreasing on $[\frac{1}{2}, 1]$.
- (B) f is strictly decreasing on $[0, \frac{1}{2}]$ and strictly increasing on $[\frac{1}{2}, 1]$.
- (C) f is strictly increasing on $[0, \frac{\sqrt{3}-1}{2}]$ and strictly decreasing on $[\frac{\sqrt{3}-1}{2}, 1]$.
- (D) f is strictly decreasing on $[0, \frac{\sqrt{3}-1}{2}]$ and strictly increasing on $[\frac{\sqrt{3}-1}{2}, 1]$.

Answer and Comments: (D). All the options involve the increasing/decreasing behaviour of $f(x)$. So the foremost method is to find $f'(x)$. If that fails (e.g. if f happens to be non-differentiable), then we have to look for other methods.

But, in order to find $f'(x)$ we must first find $f(x)$! And, in the present problem, the way $f(x)$ is defined, there is no way to apply some standard formulas to differentiate it. So, we first have to find some familiar expression for $f(x)$.

Note that for a fixed x , $f(x)$ is defined as the maximum of a function $g(y)$ of y defined by

$$g(y) = \frac{|x - y|}{x + y + 1} \quad (1)$$

for $0 \leq y \leq 1$. Because the absolute value changes its definition at 0, and x lies in the same interval as y does, we have to consider two cases depending upon whether $y \leq x$ or $y \geq x$. Doing so, (1) splits as

$$g(y) = \begin{cases} \frac{y-x}{x+y+1}, & \text{if } 0 \leq y \leq x \\ \frac{x-y}{x+y+1}, & \text{if } x \leq y \leq 1 \end{cases} \quad (2)$$

which is highly comfortable if for no reason than its familiar form. There is hardly a test at the 10+2 level where there is no problem which involves a function defined by different formulas for different subintervals. So the very sight of the big left curly bracket makes us feel that we are now on a familiar turf.

We have to maximise this for $0 \leq y \leq 1$, holding x as a constant. But since $g(y)$ has a different definitions on $[0, x]$ and $[x, 1]$, we have to separately find the maximum of $g(y)$ on each of these two subintervals. Let us call the maximum of $g(y)$ on $[0, x]$ as $M_1(x)$ and its maximum on $[x, 1]$ as $M_2(x)$. After we calculate them, the greater of the two will be the maximum of $g(y)$ on the entire interval $[0, 1]$. (This takes as much intelligence as is needed to understand that to decide the national champion in some sport, if the country is divided into two zones, then a match is played between the two zonal champions and the winner is the national champion.)

Let us first tackle $M_1(x)$, the maximum of $g(y)$ on $[0, x]$. Here $g(y) = \frac{x-y}{x+y+1}$. We can use calculus to find the maximum on $[0, x]$.

But there is a simpler way. The numerator and the denominator are both positive. But as y increases the numerator decreases (because of the negative coefficient of y) while the denominator increases. So the ratio must decrease. Hence the maximum of $g(y)$ on $[0, x]$ occurs at the left end point 0. In other words,

$$M_1(x) = g(0) = \frac{x}{x+1} \quad (3)$$

With the second interval $[x, 1]$ we are not so lucky. This time $g(y) = \frac{y-x}{x+y+1}$. Now both the numerator and the denominator increase with y and so our earlier argument breaks down. A perceptive person can still claim that the ratio is increasing because in the numerator y is clubbed with $-x$ while in the denominator it is clubbed with the bigger number $x+1$. So even though both the numerator and the denominator increase with y , the same increment in y will cause a greater proportionate increase in the numerator than the denominator. (A real life analogy would be that if the price of some commodity is hiked, between two families that use the same amounts of that commodity, the poorer family will feel the impact more.)

What if you lack such fine perceptivity? There is still a simple way out. We recast $g(y)$ by writing the numerator as $y + x + 1 - 2x - 1$. Then $g(y) = \frac{y + x + 1 - 2x - 1}{x + y + 1} = 1 - \frac{2x + 1}{x + y + 1}$. The first term is a constant. In the second term, the numerator is a constant while the denominator increases with y . So the ratio in the second term decreases as y increases. But as we are taking its negative, $g(y)$ increases. (The incorrigible lovers of calculus can differentiate to get $g'(y) = \frac{2x+1}{(x+y+1)^2}$ and observe that it is positive as $x > 0$. Like many other convenient devices such as vehicles and calculators, calculus makes you forget, and sometimes lose, some of your abilities which would have been needed in absence of those devices.)

So, no matter which method we follow, $g(y)$ is increasing on the interval $[x, 1]$. Hence

$$M_2(x) = g(1) = \frac{1-x}{x+2} \quad (4)$$

The function $f(x)$ is the larger of $M_1(x)$ and $M_2(x)$ as noted above. So,

$$f(x) = \max \left\{ \frac{x}{x+1}, \frac{1-x}{x+2} \right\} \quad (5)$$

for all $x \in [0, 1]$.

We are still a long way off. We have to decide which of the two numbers $M_1(x)$ and $M_2(x)$ is greater. The answer will depend on x . When it is not easy to directly compare two expressions, we subject them to a series of simple operations which preserve inequalities. For example, we can add a constant to both. Or we can multiply both by a positive constant. We prepare two columns one headed by $M_1(x)$ and the other by $M_2(x)$. In each column we add the two ‘descendants’ to which the comparison is reduced. We go on doing this till we reach a stage where one side can be declared greater than the other by inspection or by some other means. (The process is reminiscent of a prolonged court case which goes through hearings after hearings and finally is decided in favour of one of the parties, usually after several generations of the original combatants have passed.)

$$\begin{array}{cc} M_1(x) & M_2(x) \\ \frac{x}{x+1} & \frac{1-x}{x+2} \\ x(x+2) & (1-x)(x+1) \\ x^2 + 2x & 1 - x^2 \\ 2x^2 + 2x - 1 & 0 \end{array}$$

The justifications in each transition are usually obvious and are not specified, For example, in the very first step we multiplied both the sides by $(x+1)(x+2)$ which is positive for $x \in [0, 1]$.

We have thus reduced the problem to deciding whether the quadratic $2x^2 + 2x - 1$ is positive or negative. Its leading coefficient is positive and its roots are $\frac{-1 \pm \sqrt{3}}{2}$. We discard the negative square root as x takes only non-negative values. So, by properties of the quadratics, for $x < \frac{\sqrt{3}-1}{2}$, $M_1(x) < M_2(x)$, while for $x > \frac{\sqrt{3}-1}{2}$, $M_1(x) > M_2(x)$. For $x = \frac{\sqrt{3}-1}{2}$, $M_1(x)$ and $M_2(x)$ equal each other.

With this information, $f(x)$ now becomes

$$f(x) = \begin{cases} \frac{1-x}{x+2}, & \text{if } 0 \leq x \leq \frac{\sqrt{3}-1}{2} \\ \frac{x}{x+1}, & \text{if } \frac{\sqrt{3}-1}{2} < x \leq 1 \end{cases} \quad (6)$$

So, at long last, we have expressed $f(x)$ in a tangible, familiar form. Even this is not quite rosy, because $f(x)$ has a different definition in the two parts of the interval, one from 0 to $\frac{\sqrt{3}-1}{2}$ and the other from $\frac{\sqrt{3}-1}{2}$ to 1. But, as noted earlier, candidates taking competitive examinations at the 10+2 level feel no discomfort with such splitting. More importantly, the logic we adopted earlier to study the increasing/decreasing behaviour of $g(y)$ (for a particular constant x) can also be applied to $f(x)$, which is only a function of x . In the first subinterval $[0, \frac{\sqrt{3}-1}{2}]$, the numerator decreases while the denominator increases. So $f(x)$ is strictly decreasing on $[0, \frac{\sqrt{3}-1}{2}]$. In the second subinterval $[\frac{\sqrt{3}-1}{2}, 1]$, even though both the numerator and the denominator increase with x , the numerator increases more rapidly and so the ratio increases. (If this is still not clear to you, rewrite $\frac{x}{x+1}$ as $1 - \frac{1}{x+1}$ and then apply some brain, or, failing which, differentiate!) Hence option (D) is correct. ■

There are essentially two parts in the problem. The first is to convert the given formula for $f(x)$ to the manageable (5) and the second is to decide the increasing/decreasing behaviour of $f(x)$. The spadework needed in the first part is considerable. But the reasoning needed in it is the same as that in the second part of the problem. As a result, the problem has become very repetitious and laborious. It would have been a fair question in Paper 2 where you get 15 minutes per question. In Paper 1, you get only four minutes. So, it would have been a far better problem to give $f(x)$ as in (5) and ask to identify the intervals where it is increasing/decreasing.

Q.28 For a positive real number α , let S_α denote the set of points (x, y) satisfying

$$|x|^\alpha + |y|^\alpha = 1.$$

A positive number α is said to be *good* if the points in S_α that are closest to the origin lie only on the coordinate axes. Then

(A) all $\alpha \in (0, 1)$ are good and others are not good.

- (B) all $\alpha \in (1, 2)$ are good and others are not good.
- (C) all $\alpha > 2$ are good and others are not good.
- (D) all $\alpha > 1$ are good and others are not good.

Answer and Comments: (C). Let us first note that the adjective ‘good’ here is purely mathematical. It has very little to do with any nice or desirable qualities of the sets S_α . Unlike adjectives like ‘continuous’, ‘differentiable’, ‘increasing’ or ‘isosceles’ which are universally used, this particular adjective ‘good’ is only of a local usage. It will not be found, at least used in this sense, anywhere else in mathematics. The paper setters could as well have called it ‘bad’ or ‘green’ or anything of their choice.

The point to note is that the question is designed to test, in part, the ability to grasp a new concept and answer questions about it. Usually, in this process, the earlier knowledge we have helps. The ability to relate a new concept with what you already know is an invaluable asset and the present question deserves to be commended for testing this.

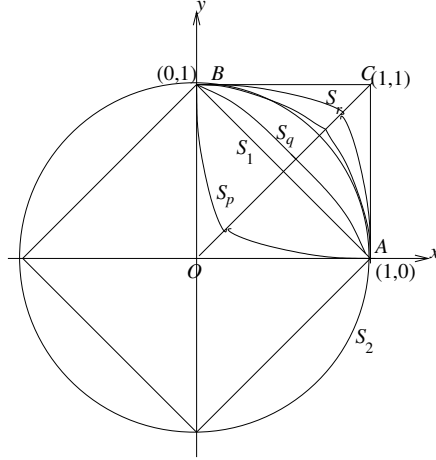
Now, coming to the problem, we note that for each $\alpha > 0$, whether a point (x, y) is in S_α or not depends only on $|x|$ and $|y|$. So, if a point (x, y) is in S_α , so are its ‘mates’ $(\pm x, \pm y)$ in the four quadrants of the plane. The same holds for the distance of a point from the origin. Hence to test if some α is good or not, it suffices to assume that (x, y) is in the first quadrant, where x, y are both non-negative and hence $|x|^\alpha + |y|^\alpha$ is the same as $x^\alpha + y^\alpha$.

Having got riddance of the absolute value, we can now identify the portions of S_α for some values of α . The easiest case is $\alpha = 1$. Here we have

$$S_1 = \{(x, y) \in \mathbb{R}^2 : x + y = 1\} \tag{1}$$

(Note that we have not specified that $x \geq 0$ and $y \geq 0$ because it is our standing hypothesis.) This is the straight line segment joining the points $(1, 0)$ and $(0, 1)$. Both these points lie on the coordinate axes and are at a distance 1 from the origin O . Clearly every point on it other than the two ends is closer to the origin than. So S_1 does not

satisfy the given condition. Hence 1 is not good. The full S_1 is shown in the figure below.



The next familiar S_α is S_2 . This is simply the circle $\{(x, y) : x^2 + y^2 = 1\}$. It is also shown in the figure. Here all points on it lie at the same distance from O . So S_2 is not good either. But the comparison of S_1 and S_2 shows that while S_1 is miserably 'bad', S_2 is only marginally so. Points on S_1 other than those on the axes are much closer to the origin than 1. In fact, the midpoint of the segment, viz. $(1/2, 1/2)$ is only at a distance $\frac{1}{\sqrt{2}}$ from O .

It is obvious that every S_α is symmetric about the line $y = x$ and also that the points $A = (1, 0)$ and $B = (0, 1)$ lie on it. Take S_p for some $0 < p < 1$. Only the first quadrant portion of it is shown in the figure. It will lie inside the triangle bounded by the axes and S_1 . So again all its interior points will be closer to O . As $p \rightarrow 0$, S_p will become more and more 'squarish' and approaches the union of the segments OA and OB . When $\alpha = q$ for some $q \in (1, 2)$, S_q will lie between S_1 and S_2 and will be bad.

When $\alpha > 2$, S_α will bulge more than S_2 . One such S_r is shown above. It is good because it lies outside the circle S_2 except for the points A and B . As $\alpha \rightarrow \infty$, S_α will approach the union of the other two sides AC and BC of the square $OACB$.

Summing up, option (C) is correct. ■

The statements made above require a justification which can be given as follows. A typical point P on S_α can be taken as $(r \cos \theta, r \sin \theta)$ where r is its distance from O and $\theta \in [0, \frac{\pi}{2}]$. For $\theta = 0$, $P = A$ while for $\theta = \frac{\pi}{2}$, $P = B$. If we exclude these values, then both $\cos \theta$ and $\sin \theta$ are positive real numbers less than 1. So their powers $\cos^\alpha \theta$ and $\sin^\alpha \theta$ decrease as α increases. Now, for P to lie on S_α we must have

$$r^\alpha (\cos^\alpha \theta + \sin^\alpha \theta) = 1 \quad (2)$$

For $\alpha = 2$, the expression in the parentheses is 1 and so $r = 1$, confirming that all points of S_2 are at a distance 1 from O . But for $\alpha > 2$, $\cos^\alpha \theta < \cos^2 \theta$ while $\sin^\alpha \theta < \sin^2 \theta$. Hence the parenthetical expression is less than 1 and therefore $r > 1$. Hence S_α is good. For $\alpha < 2$, the opposite happens and so all points on S_α other than A and B are at a distance less than 1 from O .

Although the justification needed is simple (requiring little more than the trigonometric identity $\cos^2 \theta + \sin^2 \theta = 1$ and the fact that the powers of a positive real number less than 1 decrease as the exponent increases), its articulation is unimportant. In a problem like this, one should ‘feel’ the answer from a few examples. The justification is more a ritual. So this is a very good problem and belongs to Paper 1.

In passing we mention a possible genesis of the problems. If (x, y) is a point in $\mathbb{R}^2$, by its distance from the origin, we normally understand $\sqrt{x^2 + y^2}$. Let us call this the **norm** or the **length** of the point (x, y) now treated as a vector and denote it by $\|(x, y)\|$. Then for any two points $P_1 = (x_1, y_1)$ and $P_2 = (x_2, y_2)$ in $\mathbb{R}^2$, the distance between them can be written as $\|\mathbf{v}\|$ where $\mathbf{v}$ is the difference vector $\overrightarrow{P_1 P_2} = (x_2 - x_1, y_2 - y_1)$. We can rewrite $\|(x, y)\|$ as $(x^2 + y^2)^{1/2}$ and further as $(|x|^2 + |y|^2)^{1/2}$. This may seem redundant because $|x|^2$ always equals x^2 . The reason for this rewriting is that we can generalise the concept of norm (and hence of the distance between two points of the plane) by replacing 2 by any positive real number α . So, we define the α -**norm** of (x, y) to be $(|x|^\alpha + |y|^\alpha)^{1/\alpha}$ and denote it by $\|(x, y)\|_\alpha$. The usual norm (often called the **euclidean** norm) $\|(x, y)\|$ now is simply the 2-norm, $\|(x, y)\|_2$.

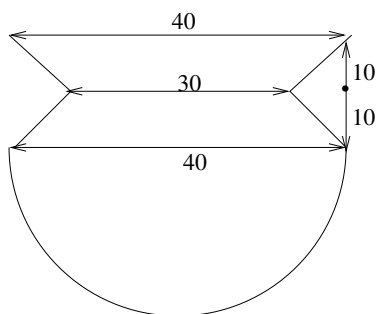
Note that as α varies, so does the α -norm. For example, if $\mathbf{v} = (1, -2)$, then $\|\mathbf{v}\|_2 = \sqrt{5}$. But $\|\mathbf{v}\|_1 = 3$, while $\|\mathbf{v}\|_{1/2} = (\sqrt{1} + \sqrt{2})^2 =$

$3 + 2\sqrt{2}$ and $\|\mathbf{v}\|_{10} = (1025)^{1/10}$ which is only slightly greater than 2. It is, in fact, possible (but not so trivial) to show that as α increases $\|\mathbf{v}\|_\alpha$ decreases for any vector $\mathbf{v} = (x, y)$ and tends to $\max\{|x|, |y|\}$ as $\alpha \rightarrow \infty$. This number is therefore denoted by $\|\mathbf{v}\|_\infty$ and called the ∞ -norm. Note that for a point P on either of the coordinate axes if we let $\mathbf{v}$ be the position vector of P , then $\|\mathbf{v}\|_\alpha$ is independent of α .

With this terminology, the set S_α in the problem is precisely the set of those points (x, y) for which $\|(x, y)\|_\alpha = 1$. Since $\|\mathbf{v}\|_\alpha$ decreases as α increases, for $\alpha > 2$, the points which are at an α -distance 1 from O will be at a greater euclidean distance from it. That is why all α 's bigger than 2 are 'good' in the terminology of the question.

One may wonder whether for $\alpha \neq 2$, the concept of the α -norm and hence also the derived concept of the α -distance between two points has any application. Certainly, in the physical world, the euclidean norm dominates all others. But there are occasions where some other distance is more relevant. It will take us too far afield to illustrate them. We only mentioned them to show that the present problem is not as weird as it may appear at first sight.

- Q.29 A water pitcher has a hemispherical bottom and a neck in the shape of two truncated cones of the same size. The vertical cross section of the pitcher with relevant dimensions is shown in the figure below.



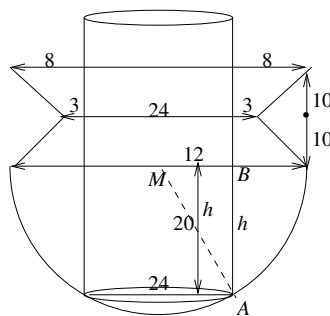
Suppose that the pitcher is filled with water to the brim. If a solid cylinder with diameter 24 cm and height greater 60 cm is inserted vertically into the pitcher as far down to the bottom as possible, how much water should remain in the pitcher?

- (A) $6316\pi \text{ cm}^3$ (B) $6116\pi \text{ cm}^3$ (C) $6336\pi \text{ cm}^3$ (D) $6136\pi \text{ cm}^3$

Answer and Comments: (A). Problems on mensuration were very common in JEE a long time ago. They were based on the formulas for the volumes of some standard solids such as cones and cylinders. Proofs of these formulas were not expected as they require calculus. Later, the JEE syllabus was revised to include calculus. After that, the mensuration problems were dropped to make room for problems testing the continuity or differentiability of some clumsily defined functions. For a working engineer, the former are far more relevant than the latter.

Mensuration can, of course, be studied under integration. In fact, finding the volumes of solids, their moments of inertia and areas of curved surfaces etc. are among the standard applications of integrals. But the integrals needed are double or triple integrals, which are too advanced for the JEE and comparable examinations. So, nowadays, the mensuration problems are relegated to physics where they are usually expected to be tackled using standard formulas.

The present problem, therefore, deserves a welcome. But the time given is inadequate. The paper setters have been kind enough to supply a diagram. But even then the very text of the problem takes more than a minute to read carefully and to figure out a way to solve it.



It is given that the cylinder is inserted vertically into the pitcher. This means that its axis is vertical and hence parallel to the axis of the pitcher. But this by itself, *does not* mean that the two axes are the same, i.e. that the axis of the cylinder passes through the centre, say M , of the hemispherical portion of the pitcher. This has to be inferred from the remaining part of the data that the cylinder is pushed as far down as possible. Although it seems obvious by common sense (and real life experience), that for this to happen, the axis must coincide

with that of the bowl, it needs a proof. We skip this proof. When the essence of the problem is mensuration, such collateral problems of geometric optimisation take a back seat. And, in any case, in MCQ's justifications hardly matter.

Now coming to the solution, the volume, say V , of the water left in the pitcher is given by

$$V = V_1 - V_2 \quad (1)$$

where V_1 is the volume of the pitcher and V_2 is the volume of the portion of the cylinder immersed into the water. We calculate both these volumes separately.

The volume of the pitcher is the sum of the volumes of its hemispherical portion and that of its neck. As all distances are in centimeters, the volumes will be in cubic centimeters. We shall drop the units cm and cm^3 in expressing the lengths and the volumes numerically. The hemisphere has radius 20 and so the volume, say H , of the hemispherical part is

$$H = \frac{2\pi}{3}(20)^3 = \frac{16000}{3}\pi \quad (2)$$

The neck is given to consist of two equal truncated cones. Each truncated cone has height 10 and the radii of its circular faces are 20 and 15. So the volume, say N , of the neck will be twice that of each truncated cone, and hence

$$\begin{aligned} N &= \frac{20\pi}{3} \times ((20)^2 + (15)^2 + 20 \times 15) \\ &= \frac{20 \times 925}{3}\pi = \frac{18500}{3}\pi \end{aligned} \quad (3)$$

Adding (2) and (3),

$$V_1 = \frac{34500}{3}\pi = 11500\pi \quad (4)$$

To calculate V_2 , we are given that the radius of the cylinder is 12. To find the volume of the portion inserted, we need to know its height. Obviously this will be less than the depth of the bowl, which is 20 +

$10 + 10 = 40$. To find out exactly what it is, we let h be the height of the portion of the cylinder lying inside the hemisphere. To find h , take any point A on the lower rim of the cylinder and let B be the point directly above it lying in the diametrical plane of the hemisphere as shown in the figure. Then $h = AB$. The triangle MBA is right angled at B . MB equals the radius of the cylinder, 12 while MA is a radius of the hemisphere and hence equals 20. Hence by the Pythagorean theorem,

$$h = \sqrt{(20)^2 - (12)^2} = 4\sqrt{5^2 - 3^2} = 4 \times 4 = 16 \quad (5)$$

(Note the time saved by taking out the common factor 4 from 20 and 12 before squaring them hastily.)

So the total height of the immersed portion of the cylinder is $16 + 20 = 36$. Hence

$$V_2 = 36\pi \times (12)^2 = 5184\pi \quad (6)$$

(Here too, some saving of time is possible by the fact that 12^3 is a familiar number 1728.)

To finish the proof we subtract (6) from (4) to get

$$V = 9500\pi - 5184\pi = 6316\pi \quad (7)$$

which tallies with (A). ■

A good problem on mensuration. But the work involved is too much, especially for those who have forgotten the formulas for the volumes of the various solids involved. The formulas for the volumes of a sphere and a cylinder are hard to forget, but that for the volume of a truncated cone (viz. $\frac{h\pi}{3}(r_1^2 + r_2^2 + r_1r_2)$ where h is the height and r_1, r_2 are the radii of the two circular faces) is likely to have been forgotten because of long non-use. Deriving it fresh by subtracting the volume of one cone from another with the same axis is time consuming.

Q.30 Let $f : [-1, 1] \longrightarrow \mathbb{R}$ be a function such that $f\left(\sin \frac{x}{2}\right) = \sin x + \cos x$ for all $x \in [-\pi, \pi]$. The value of $f\left(\frac{3}{5}\right)$ is

- (A) $\frac{24}{25}$ (B) $\frac{31}{25}$ (C) $\frac{33}{25}$ (D) $\frac{7}{5}$

Answer and Comments: (B). The function f in the problem is defined only on the interval $[-1, 1]$. The data gives its value at a point if that point is expressed as the sine of some angle. We have to find the value of $f(\frac{3}{5})$. So we have to begin by expressing $\frac{3}{5}$ as $\sin \frac{x}{2}$ for some $x \in [-\pi, \pi]$. This is a simple trigonometric equation. Since x varies in $[-\pi, \pi]$, $\frac{x}{2}$ varies in $[-\pi/2, \pi/2]$. In this interval $\alpha = \sin^{-1} \frac{3}{5}$ is uniquely defined. Moreover it lies in $(0, \pi/2)$ since $\frac{3}{5} > 0$. So $\cos \alpha$ is also positive and equals $\frac{4}{5}$.

So, we now have $x = 2\alpha$ with $\sin \alpha = \frac{3}{5}$ and $\cos \alpha = \frac{4}{5}$. Now it is only a matter of using the relation and expressing $\sin 2\alpha$ and $\cos 2\alpha$ in terms of $\sin \alpha$ and $\cos \alpha$ whose values we know. So,

$$\begin{aligned}
 f\left(\frac{3}{5}\right) &= f\left(\sin \frac{2\alpha}{2}\right) \\
 &= \sin 2\alpha + \cos 2\alpha \\
 &= 2 \sin \alpha \cos \alpha + \cos^2 \alpha - \sin^2 \alpha \\
 &= 2 \times \frac{3}{5} \times \frac{4}{5} + \frac{16}{25} - \frac{9}{25} \\
 &= \frac{24 + 7}{25} = \frac{31}{25}
 \end{aligned} \tag{1}$$

So (B) is correct. ■

An extremely simple problem about inverse trigonometric functions. Normally, when $\sin \alpha = \frac{3}{5}$, there is a tendency to simply assume that $\cos \alpha = \frac{4}{5}$ without discarding the other possibility, viz. $-\frac{4}{5}$. The data should have been chosen to penalise such a lack of scruples.

ISI BStat-BMath-UGB-2017 Paper with Comments

- Q. 1** Let the sequence $\{a_n\}_{n \geq 1}$ be defined by $a_n = \tan(n\theta)$ where $\tan \theta = 2$. Show that for all n , a_n is a rational number which can be written with an odd denominator.

Solution and Comments: Formulas which express $\tan n\theta$ as a rational function of $\tan \theta$ (i.e. as a ratio of two polynomials in $\tan \theta$) are well known for lower values of n . For example,

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} \quad (2)$$

$$\text{and } \tan 3\theta = \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} \quad (3)$$

So, one way to tackle the problem is to have a formula which expresses $\tan n\theta$ as the ratio of two polynomials in $\tan \theta$ and then put $\tan \theta = 2$. Such a formula indeed exists and is shown below.

$$\tan n\theta = \frac{\binom{n}{1} \tan \theta - \binom{n}{3} \tan^3 \theta + \dots}{1 - \binom{n}{2} \tan^2 \theta + \binom{n}{4} \tan^4 \theta + \dots}$$

This formula will give the exact value of $\tan n\theta$ if we put $\tan \theta = 2$. It is also clear that the value will be a rational number because all binomial coefficients are integers. Moreover, all the terms in the numerator and all except the first term in the denominator will be even. So the $\tan n\theta$ is a rational number which can be written with an odd denominator.

But this formula is not so well known, as its special cases above. So we give an argument which does not use it, which is possible because our goal is much less ambitious than finding the exact value of $\tan n\theta$. We proceed by induction on n . For the inductive step, we need to relate $\tan(n+1)\theta$ to $\tan \theta$.

For this we use the more basic identity (which is instrumental to derive both (1) and (2)), viz.

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \quad (4)$$

Putting $\alpha = n\theta$ and $\beta = \theta$ and then substituting $\tan \theta = 2$ we get

$$a_{n+1} = \tan(n+1)\theta = \frac{\tan n\theta + \tan \theta}{1 - \tan n\theta \tan \theta} \quad (5)$$

$$= \frac{\tan n\theta + 2}{1 - 2 \tan n\theta} \quad (6)$$

For $\tan \theta = 2$, denote $\tan k\theta$ by a_k . Evidently $a_1 = \tan \theta = \frac{2}{1}$ is a rational number with odd denominator. So the statement is true for $n = 1$. Assume that the result is true for $n = k$, i.e.

$$a_k = \tan k\theta = \frac{p}{q} \quad (7)$$

where p, q are integers and q is odd. Putting this into (5) with $n = k$, we get

$$\begin{aligned} a_{k+1} &= \frac{\frac{p}{q} + 2}{1 - 2\frac{p}{q}} \\ &= \frac{p + 2q}{q - 2p} \end{aligned} \quad (8)$$

which is a ratio of two integers and the denominator, viz. $q - 2p$ is odd because q is odd. Hence the statement is true for $n = k + 1$. So, it is true for all $n \in \mathbb{N}$. ■

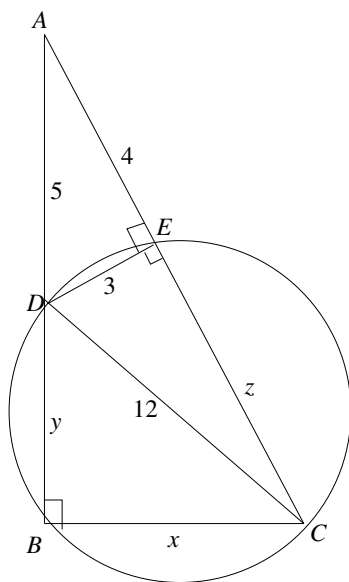
A simple problem once it is realised that we do not need to derive a formula for $\tan n\theta$ for a general θ since we are dealing with a specific angle θ and the assertion to be proved is comparatively weak.

The paper setters have also made the problem too straightforward. They could have given a slight twist by merely asking to show that $\tan(2017\theta)$ is a rational of the desired type, given that $\tan \theta = 2$. This would be perfectly in line with the modern practice of designing at least one problem in which the data involves the number of the calendar year in which the test is held. Never mind how superficial this involvement is. In fact, the candidates would first have to realise that 2017 has no role and could be replaced by any positive integer n , so that a proof by induction becomes possible.

- Q. 2 Consider a circle of radius 6. Let B, C, D and E be points on the circle such that BD and CE , when extended, intersect at A . If AD and AE have lengths 5 and 4 respectively, and DBC is a right angle, then show that the length of BC is $\frac{12 + 9\sqrt{15}}{5}$.

Solution and Comments: The solution to any good geometry problem (and geometry problems are presumed to be good unless proved otherwise!) ought to begin with a good diagram. Such a diagram is, in fact, often needed to understand the problem. And then, if it is well drawn, it might also suggest a solution. However, it is not necessary, especially when time is severely limited, that the diagram be drawn to scale. Certain vital features should not be compromised. For example, an isosceles triangle should indeed look like an isosceles one. Similarly, perpendicularity of two lines in the data should not be compromised. But, if the angle between two lines is given to be, say, 30° , it is hardly necessary to use a protractor in showing this. Similarly, some leeway can be taken as far as the lengths of some of the segments are concerned. What matters more is not so much the neatness and mathematical accuracy of the diagram as its ability to convey to you the essence of the problem, and possibly, a line of attack.

The following diagram fits this prescription.



The angle $\angle DBC$ is clearly shown as a right angle. But the segment AD which is supposed to have length 5 appears much longer in proportionate comparison with the diameter CD which is given to be 12. But that does not affect the vital calculations.

Now, coming to the solution, the problem asks us to show that the length of BC is $\frac{12+9\sqrt{15}}{5}$. If instead of this horrible number, we had some simple number, say 4, then one way to solve the problem would reduce to showing that $BC = AE$ and then we can look for some intermediary (usually not given in the figure) which equals BC on one hand and AE on the other. The search for such intermediary would make the problem interesting. That is usually the beauty of many geometry problems. Things which look unrelated on the face of it, turn out to be closely related through such an intermediary. The excitement is akin to that in discovering that the total stranger who sits next to you on a flight turns out to be your cobrother's brother!

The present problem is not meant for such excitement. The figure $\frac{12+9\sqrt{15}}{5}$ will have to be derived through some hard calculation, possibly involving a quadratic equation. In such a problem, it is best to introduce symbols for the lengths of the various line segments (other than those whose lengths are given) and reduce the problem to solving a system of equations. We begin by calling the sides BC, BD and CE as x, y, z respectively. Our interest is only in x . But we need other variables as auxiliary variables. Generally, one should minimise the number of such auxiliary variables. For example, we could introduce one more variable, say w , for the length of the segment DE . But that can be obviated if we observe that since $DBCE$ is a cyclic quadrilateral and $\angle DBC$ is a right angle, $\angle DEC$ is also a right angle. But then so is $\angle AED$ and so by Pythagoras theorem we get $DE = 3$.

Thus we have three unknowns x, y, z and to determine them we need a system of three (independent) equations. These are easily obtained from the right angled triangles $\triangle DBC, \triangle DEC$ and $\triangle ABC$. Keeping in mind that CD is a diameter and hence has length 12, we get,

$$x^2 + y^2 = 144 \quad (1)$$

$$z^2 + 9 = 144 \quad (2)$$

$$\text{and } x^2 + (y + 5)^2 = (z + 4)^2 \quad (3)$$

From this point onwards, this is purely a problem of algebra. Using (1) and (2), (3) can be simplified to give

$$10y + 25 = 8z + 7 \quad (4)$$

i.e.

$$5y + 9 = 4z \quad (5)$$

Substituting from (1) and (2) into this we get

$$5\sqrt{144 - x^2} + 9 = 4\sqrt{135} = 12\sqrt{15} \quad (6)$$

So, we have finally got an equation in x , in whose value we are interested. Simplifying and squaring, we have

$$144 - x^2 = \frac{(12\sqrt{15} - 9)^2}{25} \quad (7)$$

and hence

$$\begin{aligned} x^2 &= 144 - \frac{(12\sqrt{15} - 9)^2}{25} \\ &= \frac{144 \times 25 - 144 \times 15 - 81 + 216\sqrt{15}}{25} \end{aligned} \quad (8)$$

$$= \frac{1440 - 81 + 216\sqrt{15}}{25} = \frac{1359 + 216\sqrt{15}}{25} \quad (9)$$

Note that in (8), we have left the products 144×25 and 144×15 as they are. This helps in deriving (9) quickly. Minor as they are, such savings have some elegance. (In the present case, the saving is based on the distributivity of multiplication over subtraction.)

To finish the solution, we merely need to show that the numerator in (9) equals $(12 + 9\sqrt{15})^2$. Here also a minor saving of time is possible by taking out 3 as a common factor from 12 and 9. Then

$$(12 + 9\sqrt{15})^2 = 9(4 + 3\sqrt{15})^2 = 9(151 + 24\sqrt{15}) = 1359 + 216\sqrt{15}$$

which completes the solution. ■

In the solution above, the results needed from geometry were only the Pythagoras theorem (which was used four times) and a very elementary

property of cyclic quadrilaterals, viz. that their opposite angles add to 180° . Those who know a little more about circles can get an easier derivation of (5). The result needed is that for any point A outside a circle, if a line through A cuts the circle at points X and Y , then the product $AX \cdot AY$ is independent of the line. This constant, is called the **power** of that point w.r.t. that circle. In the present problem, we can apply it to the lines ADB and AEC to get

$$5(y + 5) = 4(z + 4) \quad (10)$$

which implies (5). The proof of this result is based on the fact that an external angle of a cyclic quadrilateral equals the opposite angle. As a result, the triangles $\triangle AED$ and $\triangle ABC$ are similar to each other. Equating the ratios of the corresponding sides we get (10).

Whichever way look at it, this is a simple geometric problem. But the computations involved hijack it to the domain of algebra. So, this is a geometric problem which is not ‘so’ good as a ‘purely’ pure geometry problem!

Q. 3 Suppose $f : \mathbb{R} \longrightarrow \mathbb{R}$ is a function given by

$$f(x) = \begin{cases} 1 & \text{if } x = 1 \\ e^{(x^{10}-1)} + (x-1)^2 \sin \frac{1}{x-1} & \text{if } x \neq 1 \end{cases}$$

(a) Find $f'(1)$

(b) Evaluate $\lim_{u \rightarrow \infty} \left[100u - u \sum_{k=1}^{100} f\left(1 + \frac{k}{u}\right) \right]$.

Solution and Comments: (a) is easy. The second term is differentiable at 1 (if we set its value equal to 0 at 1) with derivative 0 because for $x \neq 1$,

$$\frac{(x-1)^2 \sin \frac{1}{x-1} - 0}{x-1} = (x-1) \sin \frac{1}{x-1} \quad (1)$$

which tends to 0 as x tends to 1 because the first factor tends to 0 while the second one is bounded. So the problem reduces to finding the derivative of the first term, viz., $e^{(x^{10}-1)}$ at $x = 1$. By the chain rule this comes out as the value of $e^{(x^{10}-1)} 10x^9$ at 1. So $f'(1) = 10$.

For (b), we first recast the expression whose limit is to be taken to get

$$\begin{aligned} \lim_{u \rightarrow \infty} \left[100u - u \sum_{k=1}^{100} f \left(1 + \frac{k}{u} \right) \right] &= \lim_{u \rightarrow \infty} u \sum_{k=1}^{100} 1 - f \left(1 + \frac{k}{u} \right) \\ &= \sum_{k=1}^{100} \lim_{u \rightarrow \infty} u \left[f(1) - f \left(1 + \frac{k}{u} \right) \right] \quad (2) \\ &= \sum_{k=1}^{100} k \lim_{u \rightarrow \infty} \frac{f(1) - f \left(1 + \frac{k}{u} \right)}{k/u} \quad (3) \end{aligned}$$

where (2) is obtained by moving the limit inside a summation, which is valid since only a finite sum is involved.

The problem is now reduced to finding, for each k from 1 to 100, the limit in (3) and adding these limits. From the nature of the ratio appearing in (3), it is tempting to apply Lagrange's Mean Value Theorem and to write the ratio as $-f'(c)$ for some c in the interval $(1, 1 + \frac{k}{u})$. The trouble is that this intermediate point c will depend not only on k but on u as well. It might be argued that such variation does not matter because we are only interested in the limit of the expression as $u \rightarrow \infty$ and no matter what c is, as long as it lies in the interval $(1, 1 + \frac{k}{u})$, it will tend to 1 from the right as $u \rightarrow \infty$. So the limit in (3) equals $\lim_{c \rightarrow 1^+} -f'(c)$ and hence it is $-f'(1)$ i.e. -10 , for every k .

The catch here is that this argument will require the continuity of $f'(x)$ at $x = 1$. But because of the second term, viz., $(x-1)^2 \sin(\frac{1}{x-1})$, in the expression for $f(x)$, although f is differentiable at 1, the derivative is not continuous at 1. (It is helpful to recall here that if we write y for $x-1$, then the hierarchy $\sin \frac{1}{y}, y \sin \frac{1}{y}, y^2 \sin \frac{1}{y}, y^3 \sin \frac{1}{y}, y^4 \sin \frac{1}{y}, \dots$ of functions (all of which are set to 0 at $y = 0$) is a well known example of the progressively improved behaviour at 0. The first function is not even continuous at 0, the second one is continuous but not differentiable at 0, the third one (which is involved in the present problem) is differentiable but not continuously differentiable at 0, the fourth one is continuously differentiable at 0 but fails to have a second derivative at 0 and so on.)

It is tempting to try to salvage the situation using the all popular l'Hôpital's rule. For a fixed k , put $y = \frac{k}{u}$. Then $y \rightarrow 0^+$ as $u \rightarrow \infty$

and so the limit in (3) is the same as $\lim_{y \rightarrow 0^+} \frac{f(1) - f(y)}{y}$. By l'Hôpital's rule, this equals $\lim_{y \rightarrow 0^+} \frac{-f'(1+y)}{1}$. But that does not help because as noted above, even though $f'(1)$ exists, $\lim_{y \rightarrow 0^+} f'(1+y)$ does not exist. So, l'Hôpital's rule is not of much help either.

Fortunately, there is a much easier way out. Putting $y = \frac{k}{u}$ as above, the limit in (3) equals $-f'(1)$ *from the very definition of a derivative*. No fancy theorems are needed for it. As we already calculated $f'(1)$ as 10 in (a), the sum in (3) is simply $-10 \sum_{k=1}^{100} k$ which comes out to be $-500 \times 101 = -50500$. ■

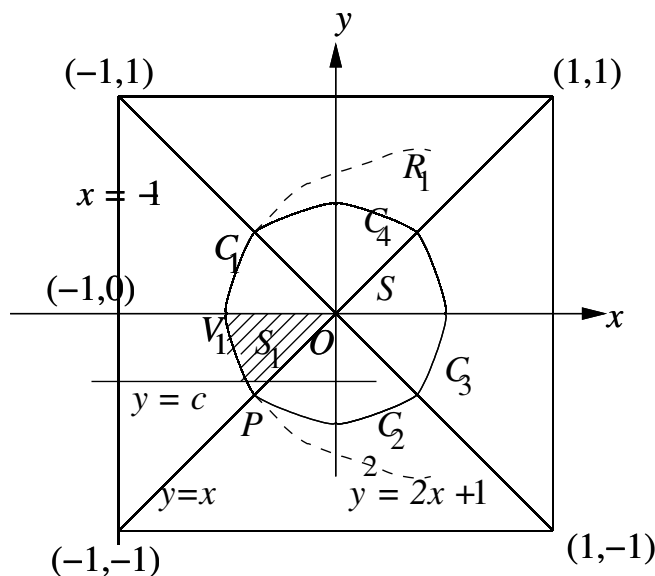
The paper setters have done a wise thing by not asking this problem in Paper 1. It does not take much time to arrive at the correct answer. Nor is the correct justification very subtle. The problem is more a test of the ability to realise how a tempting justification will not apply and it is only in Paper 2 that this ability can be tested.

- Q. 4 Let S be a square formed by the four vertices $(1, 1)$, $(1, -1)$, $(-1, 1)$ and $(-1, -1)$. Let the region R be the set of points inside S which are closer to the center than any of the four sides. Find the area of the region R .

Solution and Comments: This problem is a straight replica of a 1995 JEE problem. The answer comes out to be $\frac{16\sqrt{2} - 20}{3}$. It is obtained by dividing the region into eight mutually congruent subregions each of which is bounded by a pair of straight lines and a parabola. The reason the parabolas enter the picture is that each parabola is the locus of a point which is equidistant from a fixed point (the focus) and a fixed line, the directrix. So, for a point on one side of the parabola the focus is closer than the directrix while for a point on the other side of the parabola, the directrix is closer than the focus. In the present problem, we have to deal with four parabolas all having their foci at the origin O but their directrices are the four sides of the square.

For the full solution see Comment No. 4 of Chapter 17. (There is a minor difference of notation. The region there is denoted by S and

not R . We reproduce here the diagram in the solution.) ■



Although the problem is mathematically a replica of the JEE 1995 problem, it is probably picked from some American source as suspected from the spelling ‘center’ rather than ‘centre’. The latter is used consistently elsewhere.

Q. 5. Let $g : \mathbb{N} \rightarrow \mathbb{Z}$ with $g(n)$ being the product of the digits in n .

- (a) Prove that $g(n) \leq n$ for all $n \in \mathbb{N}$
- (b) Find all n for which $n^2 - 12n + 36 = g(n)$

Solution and Comments: The digits of a number n depend on the base chosen. Usually, we take the base as 10. But this should have been specified in the definition of $g(n)$. Also it is to be assumed that there are no leading zeros as otherwise $g(n)$ will not be well defined. The function g is not a frequently occurring function. When a new function is defined for a problem, it is customary to illustrate it with one or two examples. Trivially, $g(n) = 0$ if even one digit of n is 0. Also the $g(n)$ remains the same if the digits are reshuffled in any manner. Thus $g(1224) = g(2142) = 16$.

Trivially, if n is a single digit number i.e. a number from 1 to 9, then $g(n) = n$. Beyond 9, $g(n)$ cannot grow as fast as n , because multiples of 10 will have to be moved to the 10's place where they will be only single digits. But the digit in the unit's place is at most 9. So if n is a two digit number, say ab , then $n = 10a + b$. But $g(n) = a \times b \leq 9a$. Hence $g(n) < n$.

A similar argument applies for any number with more than one digit. Let the digits of n , read from the right (i.e. the unit's place) be $x_0, x_1, x_2, \dots, x_r$ where $x_r \neq 0$. We may call $r + 1$ the 'length' of n . (Actually, r is the integral part of $\log_{10} n$. But that is not so vital here.) We apply induction on r . For $r = 0$, n is a single digit number x_0 and both $g(n)$ and n equal x_0 . Suppose that the assertion is proved for all numbers whose length is $k + 1$. Now suppose n is a number with $k + 2$ digits $x_0, x_1, \dots, x_k$ and x_{k+1} , read from the right to left. Let m be the number whose digits are $x_1, x_2, \dots, x_{k+1}$, again read backwards from the unit's place digit which is now x_1 . (For example, if $n = 2337$ then m is simply 233.) Clearly,

$$n = 10m + x_0 \quad (1)$$

Since m has length $k + 1$, by the induction hypothesis, we have

$$g(m) = x_1 \times x_2 \times \dots \times x_{k+1} \leq m \quad (2)$$

But $g(n)$ is simply $x_0 g(m)$ and x_0 is at most 9. So,

$$g(n) = x_0 g(m) \leq 9g(m) \leq 9m \quad (3)$$

Putting (1) and (3) together,

$$g(n) \leq 9m < 10m + x_0 \quad (4)$$

So the assertion is true for all numbers of length $k + 2$. Hence by induction, it holds for all n . (In fact, strict inequality holds for all numbers except single digit ones.) ■

(b) If $n^2 - 12n + 36 = g(n)$, then combining this with the inequality in (a), we have

$$n^2 - 13n + 36 \leq 0 \quad (5)$$

Factorising the L.H.S.,

$$(n - 4)(n - 9) \leq 0 \quad (6)$$

which is possible only when $4 \leq n \leq 9$, i.e. for $n = 4, 5, 6, 7, 8$ and 9 . In all these cases, $g(n) = n$. But since $n^2 - 12n + 36$ is also a perfect square, for it to equal $g(n)$ and hence n , n must also be a perfect square. Hence the only possibilities are $n = 4$ and $n = 9$. By actual substitution, $(n - 6)^2$ equals 4 and 9 respectively for these values. So (b) holds only for $n = 4$ and $n = 9$. ■

Part (b) is a trivial consequence of (a). It is probably meant only to give some consolation credit to candidates who cannot prove (a), but can derive (b) from it. The crux of the problem is (a). Even this is fairly easy once you do some experimentation with small values of n . So, this problem is a good test of a candidate's ability to understand a new concept, do some experimentation and get the clue to the right answer (in the present problem, induction on the number of digits). These desirable qualities do not come automatically with drill work. There are candidates who have huge repertoires of solved problems. Ask them to prove a trigonometric identity about triangles and they will immediately tell you which standard identity to start with. Similarly, they know how to pull the right trick to effortlessly evaluate some integral. But ask them to think fresh and they draw a blank. So a few problems of the present type are welcome and Paper 2 is the right setting for them.

- Q. 6 Let p_1, p_2, p_3 be primes with $p_2 \neq p_3$ such that $4 + p_1p_2$ and $4 + p_1p_3$ are perfect squares. Find all possible values of p_1, p_2, p_3 .

Solution and Comments: It is not given that p_1, p_2, p_3 are odd. But this follows from the rest of the data. The only even prime is 2. So if one of the three primes is even then at least one of the numbers $4 + p_1p_2$ and $4 + p_1p_3$ will be either 8 (if two of the three primes are 2) or be of the form $4 + 2k$ for some odd integer k . 8 is not a perfect square. Nor is a number of the form $4 + 2k$ with k odd because it is divisible by 2 but not by 4.

So, we assume that p_1, p_2, p_3 are all odd. It is given that $p_2 \neq p_3$. We claim that they are all distinct. For otherwise, $p_1 = p_2$ (say). But then

$4 + p_1^2 = u^2$ which gives $p_1^2 = (u + 2)(u - 2)$. By uniqueness of prime factorisation, p_1 will equal both $u + 2$ and $u - 2$, which is impossible.

Thus we assume that p_1, p_2, p_3 are three distinct odd primes. Since $4 + p_1 p_2 = u^2$ for some u , we once again have $p_1 p_2 = (u + 2)(u - 2)$. Hence the smaller of p_1 and p_2 equals $u - 2$ and the other $u + 2$. So the primes p_1 and p_2 differ by 4. By a similar reasoning, p_1 and p_3 differ by 4. Since p_2 and p_3 are distinct, p_1 must be the smaller member of one pair and the larger one of the other. Without loss of generality, we may suppose that $p_2 < p_1 < p_3$. So the problem reduces to finding all triplets (p_2, p_1, p_3) of primes which form an A.P. with common difference 4. All primes greater than 3 are of the form $6k + 1$ or $6k - 1$. The only way p_1 and p_3 can differ by 4 is if $p_1 = 6k + 1$ and $p_3 = 6k + 5$ for some integer k . But then p_2 has to be $6k + 1 - 4$ i.e. $6k - 3$, which is a prime only for $k = 1$.

Thus the only possible solutions are $p_1 = 7$ and $p_2 = 3, p_3 = 11$ (or vice versa). That these primes actually give solutions is seen from $4 + 7 \times 3 = 25 = 5^2$ and $4 + 7 \times 11 = 81 = 9^2$. ■

A good, simple problem based on uniqueness of prime factorisation and the fact that all odd primes greater than 3 are of the form $6k \pm 1$. Initially, the candidate has to do some work to rule out the degenerate cases where one of the primes is 2 or where two of them are equal. This work is not relevant to the main theme of the problem. Still, in a test where candidates have to justify their answers, a scrupulous candidate may spend time writing the justification, lest he loses credit. If the paper setters do not really intend to test this, he would be made a fool of. In that case it would have been better to give p_1, p_2, p_3 as three distinct odd primes right in the statement of the question.

Q. 7 Let $A = \{1, 2, \dots, n\}$. For a permutation $P = (P(1), P(2), \dots, P(n))$, of the elements of A , let $P(1)$ denote the first element of P . Find the number of all such permutations P so that for all $i, j \in A$:

- (a) if $i < j < P(1)$, then j appears before i in P ; and
- (b) if $P(1) < i < j$, then i appears before j in P .

Solution and Comments: In such problems, it pays to work out at

least one special case by hand, for a small value of n . If this value of n is too small like 1 or 2, the work might not lead to any clue for the general case. On the other hand, too large a value would entail a lot of work just for trial. Let us take a reasonably small value, say $n = 7$. So we have a permutation P of the symbols 1, 2, 3, 4, 5, 6 and 7. In all there are $7!$ permutations. But we have to count only those which satisfy the conditions (a) and (b). So, let us fix the first element of P as, say 5. (Again, the choice of the extreme values 1 and 7 should be avoided as they may not be sufficiently representative. Similarly $P(1) = 4$ is ruled out on the ground that the symmetric location of 4 between the extremes 1 and 7 may lead to some simplifications which will not hold in general.)

So, suppose $P = (5, a, b, c, d, e, f)$ is a permutation in which a, b, c, d, e, f are integers from 1 to 7, other than 5. Condition (a) means that all the integers less than 5 (i.e. the integers 1, 2, 3 and 4) must appear in a descending order in the sequence (a, b, c, d, e, f) while (b) means that the integers 6 and 7 must appear in an ascending order in this sequence. Once this is realised, it is clear that our freedom is over the moment we fix the four places among a, b, c, d, e, f which the integers from 1 to 4 should occupy, because once they are fixed, we must send 1, 2, 3, 4 to them in a descending order and now only two places are left which will have to be filled by 6 and 7 in an ascending order.

The net result is that with $n = 7$ and $P(1) = 5$, the number of desired permutations is $\binom{6}{4}$. The same reasoning applies if $P(1)$ has any other value from 1 to 7. (In the extreme case where $P(1) = 1$, the integers 2 to 7 must appear in the descending order, viz. as 7, 6, 5, 4, 3 and 2 and this is consistent with $\binom{6}{0} = 1$. Therefore the total number of permutations of the desired type for $n = 7$ is

$$\sum_{k=1}^7 \binom{6}{k-1} \tag{1}$$

With a change of index, this sum is simply $\sum_{r=0}^6 \binom{6}{r}$. It is well known

that this sum is 2^6 . (One way to see this is to put $x = 1$ in the binomial expansion of $(1 + x)^6$.)

We hardly need to give the details elaborately for the general case. For each $m = 1, 2, \dots, n$, the number of permutations of the desired type for which $P(1) = m$ is $\binom{n-1}{m-1}$. Hence the total number of such permutations is the sum $\sum_{m=1}^n \binom{n-1}{m-1}$. With a change of index and the binomial theorem, this comes out to be 2^{n-1} . ■

Another good problem testing the ability to understand, focus on the essentials, do experimentation and get a clue. The problem is similar to another problem given below.

Find the number of permutations of the set $\{1, 2, \dots, n\}$ in which every integer except the one appearing last, is followed, although not necessarily immediately followed, by another integer which differs from it by one.

In this problem, if P is a desired permutation, then all the integers less than $P(n)$ must occur in an ascending order, while those greater than $P(n)$ must occur in a descending order. So we get the same answer as above, viz. 2^{n-1} . But in this problem it takes some time to get this realisation. The reasoning is as follow. Suppose $P(n) = r$. If $r > 1$, then 1 appears somewhere before r . But 2 must not appear before 1 because it is the only 'neighbour' 1 has. So, 1 is followed by 2. Now 3 must appear later because it is the only neighbour of 2 which is now left (the other neighbour 1 having appeared before 2). This will go on till r . For integers bigger than r we start from the highest one, viz. n which has only one neighbour, viz. $n - 1$. So, $n - 1$ must come after n . But then it has only one neighbour, viz. $n - 2$ left to follow it. Continuing like this, integers bigger than r must occur in a descending order.

The present problem is of a similar spirit but simpler. Hence it is a reasonable question to ask.

Q. 8 Let k, n and r be positive integers.

(a) Let $Q(x) = x^k + a_1x^{k+1} + \dots + a_nx^{k+n}$ be a polynomial with real

coefficients. Show that the function $\frac{Q(x)}{x^k}$ is strictly positive for all real x satisfying

$$0 < |x| < \frac{1}{1 + \sum_{i=1}^n |a_i|}$$

- (b) Let $P(x) = b_0 + b_1x + \dots + b_rx^r$ be a non zero polynomial with real coefficients. Let m be the smallest number such that $b_m \neq 0$. Prove that the graph of $y = P(x)$ cuts the x -axis at the origin (i.e., P changes signs at $x = 0$) if and only if m is an odd integer.

Solution and Comments: (a). Denote $\frac{Q(x)}{x^k}$ by $f(x)$. For $x \neq 0$, $f(x)$ is simply the polynomial

$$f(x) = 1 + a_1x + a_2x^2 + \dots + a_nx^n \quad (2)$$

$f(0)$ is not defined. But if we set $f(0) = 1$ then $f(x)$ is continuous at 0. Since $f(0) = 1 > 0$, by continuity, $f(x) > 0$ for all x in some neighbourhood $(-\delta, \delta)$ of 0. But the problem asks us to show that this is the case for a very specific δ , viz.

$$\delta = \frac{1}{1 + \sum_{i=1}^n |a_i|} \quad (3)$$

For this we need to analyse $f(x)$ closely. The δ given to us is the reciprocal of $1 + \sum_{i=1}^n |a_i|$ and so the condition $|x| < \delta$ is equivalent to

$|\frac{1}{x}| > 1 + \sum_{i=1}^n |a_i|$. This suggests that it would be more convenient to

make a substitution $x = \frac{1}{y}$ to express $f(x)$ in terms of some function, say $g(y)$, of y and then show that $g(y)$ is positive for all y for which $|y| > 1 + \sum_{i=1}^n |a_i|$.

The first part is easy. Put $y = \frac{1}{x}$ which is valid for $x \neq 0$. Then

$$f(x) = \frac{y^n + a_1y^{n-1} + a_2y^{n-2} + \dots + a_{n-1}y + a_n}{y^n} \quad (4)$$

The numerator is a monic polynomial of degree n , say $g(y)$, in y . So, for all $y \neq 0$,

$$f(x) = \frac{g(y)}{y^n} \quad (5)$$

We first assume $x > 0$ and hence $y > 0$. Then the denominator y^n is positive and so proving $f(x) > 0$ for $0 < x < \delta$ is equivalent to proving that

$$g(y) = y^n + a_1 y^{n-1} + a_2 y^{n-2} + \dots + a_{n-1} y + a_n > 0 \quad (6)$$

whenever,

$$y > 1 + \sum_{i=1}^n |a_i| \quad (7)$$

We are not given anything about the signs of the coefficients $a_0, a_1, \dots, a_{n-1}, a_n$. The positive coefficients will cause no problem. So, it suffices to consider only those terms $a_i y^{n-i}$ for which $a_i < 0$. But we shall give a proof which is independent of the signs.

Note first that (7) ensures that $y > 1$ and hence $y^r < y^s$ whenever $r < s$. Now since $y > 1 + |a_1| + |a_2| + \dots + |a_n|$, we have

$$\begin{aligned} y^n &= y y^{n-1} \\ &> (1 + |a_1| + |a_2| + \dots + |a_n|) y^{n-1} \\ &= y^{n-1} + |a_1| y^{n-1} + |a_2| y^{n-1} + \dots + |a_{n-1}| y^{n-1} + |a_n| y^{n-1} \quad (8) \\ &> |a_1| y^{n-1} + |a_2| y^{n-2} + |a_3| y^{n-3} + \dots + |a_{n-1}| y + |a_n| \quad (9) \end{aligned}$$

where in going from (8) to (9), we have dropped the first term y^{n-1} which is positive and from the third term onwards, replaced higher powers of y by lower ones.

(9) gives a lower bound on the first term of the R.H.S. of (6). Keeping the other terms as they are, we get

$$\begin{aligned} g(y) &> (a_1 + |a_1|) y^{n-1} + (a_2 + |a_2|) y^{n-2} + \dots \\ &\quad + (a_{n-1} + |a_{n-1}|) y + (a_n + |a_n|) \end{aligned} \quad (10)$$

whenever (7) holds. Now, regardless of the sign of a_i , the expression $a_i + |a_i|$ and hence also the term $(a_i + |a_i|) y^{n-i}$ is non-negative. As this

holds for every term of (10), we get that $g(y) > 0$ for all $y > 1 + \sum_{i=1}^n |a_i|$.

Going back to x we have proved that

$$f(x) = \frac{Q(x)}{x^k} > 0 \quad (11)$$

for all real x satisfying $0 < x < \delta$. We still need to prove this for $-\delta < x < 0$. We can again put $y = \frac{1}{x}$ and argue as above. But now y will be negative and so we shall have to be more careful in handling inequalities involving powers of y . There is a way to avoid all this by applying what we have already done to a different polynomial. In (2) we replace x by $-x$. This gives a new polynomial in x , say $h(x)$, defined by

$$h(x) = 1 + b_1x + b_2x^2 + \dots + b_nx^n \quad (12)$$

where $b_i = (-1)^i$ for $i = 1, 2, \dots, n$. Still, $|b_i| = |a_i|$ for all i and so

$$\delta = \frac{1}{1 + \sum_{i=1}^n |b_i|} \quad (13)$$

If $-\delta < x < 0$, then $0 < -x < \delta$ and so by applying our work to $h(x)$ we have

$$h(-x) > 0 \quad (14)$$

But $h(-x)$ is the same as $f(x)$. So we get that $f(x) > 0$ for $-\delta < x < 0$ too. This completes the proof in all cases. ■

(b) This is of the same spirit as (a) but now we are more relaxed because we merely have to see what happens in a sufficiently small neighbourhood of 0. The size of this neighbourhood is immaterial as long as it is positive. The coefficients $b_0, b_1, \dots, b_{m-1}$ all vanish. But $b_m \neq 0$. So, we write

$$P(x) = b_mx^m(1 + a_1x + a_2x^2 + \dots + a_{r-m}x^{r-m}) \quad (15)$$

where $a_i = \frac{b_{m+i}}{b_m}$ for $i = 1, 2, \dots, r-m$. The last factor is positive at $x = 0$ and hence, by continuity, for all x in some neighbourhood of

0. So, the sign of $P(x)$ in this neighbourhood is the same as that of b_mx^m . b_m is a constant. If m is even, then $x^m > 0$ for all x in this neighbourhood (except at 0) and hence $P(x)$ maintains its sign as x passes through 0. Depending upon the sign of b_m , $P(x)$ will be either positive on both the sides of 0 in this neighbourhood or negative on both the sides. But if m is odd, then $P(x)$ will have opposite signs on the two sides. ■

Part (b) is absolutely trivial as compared with (a) and does not need (a) for its solution. One fails to see the purpose behind asking it. Perhaps, as in Q. 6, the idea is to award some consolation credit to those who can do (b) but not (a).

But Part (a) deserves some comments. In essence, it shows that all the real roots of the monic polynomial $y^n + a_1y^{n-1} + \dots + a_{n-1}y + a_n$ lie in the interval $(-M, M)$ where $M = 1 + \sum_{i=1}^n |a_i|$. This upper bound on the size of a real root of a monic polynomial is due to **Lagrange**. As noted in the solution, we could have as well dropped the terms with positive coefficients. In fact, one can concentrate on the ‘worst’ term, i.e. one for which a_i is negative and has the largest absolute value. This way we get sharper upper bounds. Considerable work has been done in cornering the roots of a polynomial in as small as intervals as possible, because that is the next best thing to do when one cannot obtain the roots exactly. (And for polynomials of degree 5 or above, there is no formula to express the roots in terms of the radicals of expressions involving the coefficients.) The quadratic formula is exceptional and deceptive.

CONCLUDING REMARKS

As is to be expected, the entire test is a mixed bag. Still, Paper 1 is much better than Paper 1 of 2015 which was highly disappointing.

Unlike in the past three years, there is no question which stands out as the ‘man of the match’. If a selection has to be made there will be a tie between Q.23 of Paper 1 (about restricted permutations of the letters of the English alphabet) and Q.8 of Paper 2 (based on Lagrange’s upper bound on the absolute values of the roots of a polynomial).

Some areas such as mensuration have appeared after a long gap (Q.29 of Paper 1). Mercifully, there is no question regarding the integral part function!. Of course, there are at least three questions which involve the other favourite of paper setters, viz. the absolute value function (Q.3, Q.27 and Q.28). The first one can be answered in a sneaky manner. Q.27 is laborious, because even though the essential idea is simple, it has to be applied again and again. Q.28 is a good question testing a candidate’s ability to grasp a fresh concept. The same thing can be said about Q.5 in Paper 2.

Overall, geometry has gotten a large representation. Q.2, Q.4, Q.8, Q.10, Q.12, Q.15 Q.17 and Q.24 of Paper 1 and Q.2 of Paper 2 are all based on pure geometry, sometimes coupled with something extra such as trigonometry or complex numbers. There is only one locus problem (Q.13 of Paper 1). It is different from the ordinary locus problems in that the choice of two parameters is more advantageous than one. Q.21 in Paper 1 (about finding an angle of a triangle given two trigonometric equations) is also interesting.

Comparatively, number theory has got a beating. Q.6 of Paper 1 involves absolutely nothing more than finding the g.c.d. of three numbers. The only other number theory problem (Q.6 in Paper 2) about three primes is also way too elementary.

Combinatorics has fared better. Q.14 of Paper 1 about finding the number of two-to-one functions is far better than Q.2 of ISI 2015 which was absolutely trivial. Both the problems on permutations are good, the better one (Q.23 of Paper 1) was already mentioned.

There are three problems about finding areas. Two of them (Q.9 from Paper 1 and Q.4 from Paper 2) require integration. Both are good problems. But it is shocking that they are direct replicas of old JEE problems. This is really ironic. At the end of the JEE 2017 commentary, ISI 2016 paper was mentioned with approval as a model JEE can emulate. But now we have two earlier JEE problems appearing in 2017 ISI.

So, it is now difficult to rank one test as consistently better than the other. Let us hope that in the coming years, there will be a healthy competition between JEE and ISI, each trying to come up with some original problems.